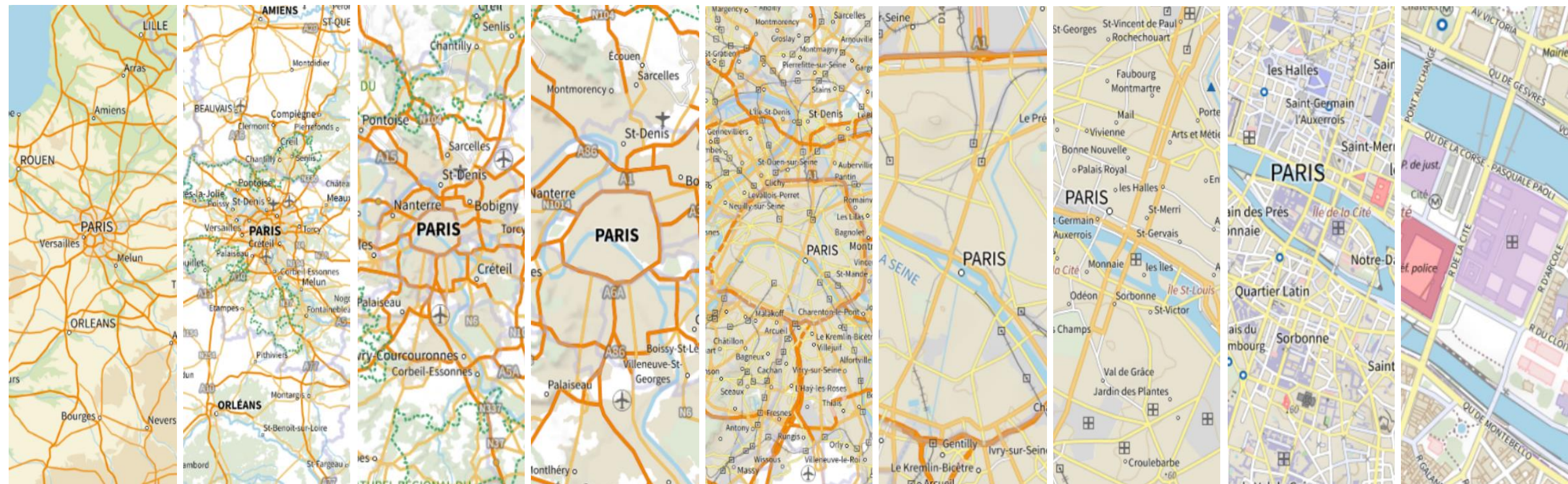


La progressivité en cartographie

Bérénice Le Mao & Guillaume Touya

Journée de la Recherche - 2025



Est-ce que c'est progressif ?

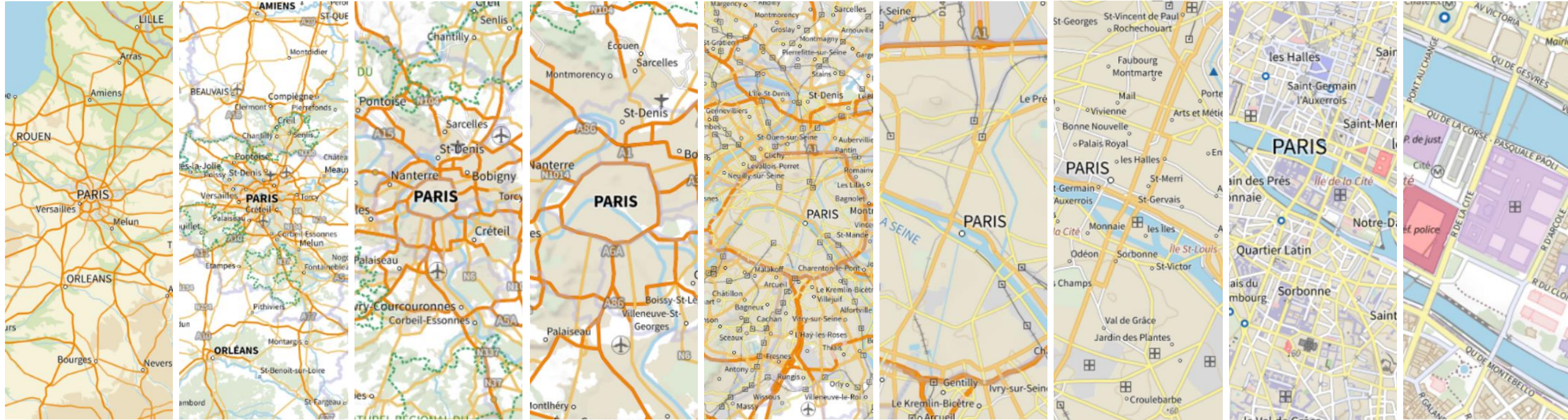


Plan IGN Classique



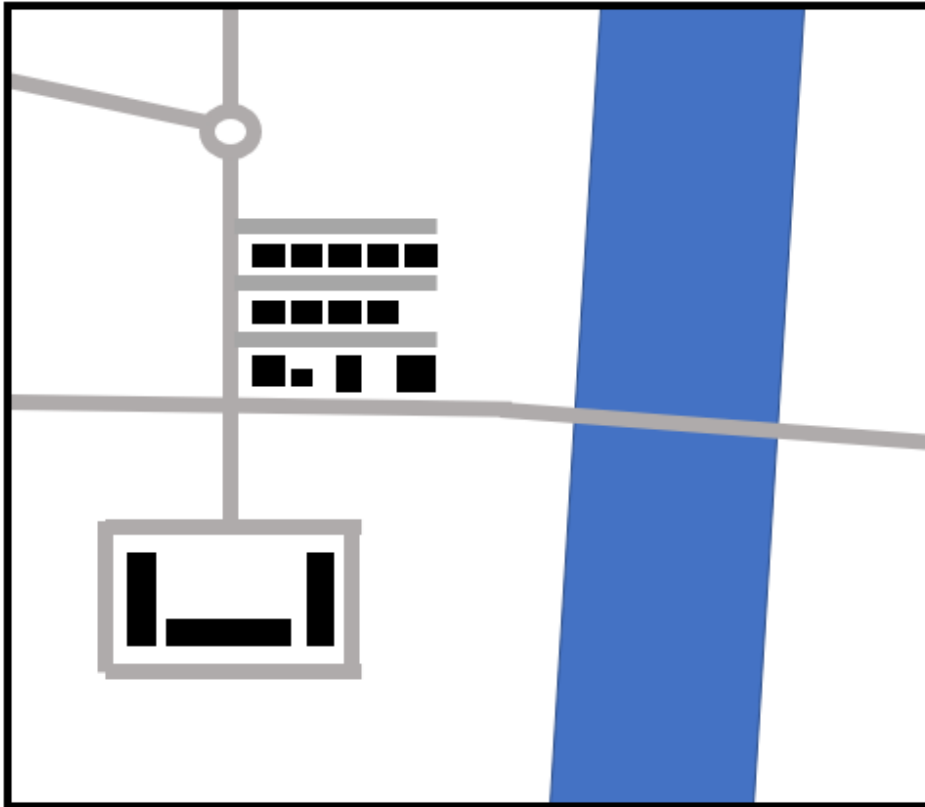
Série de cartes individuelles

Progressivité

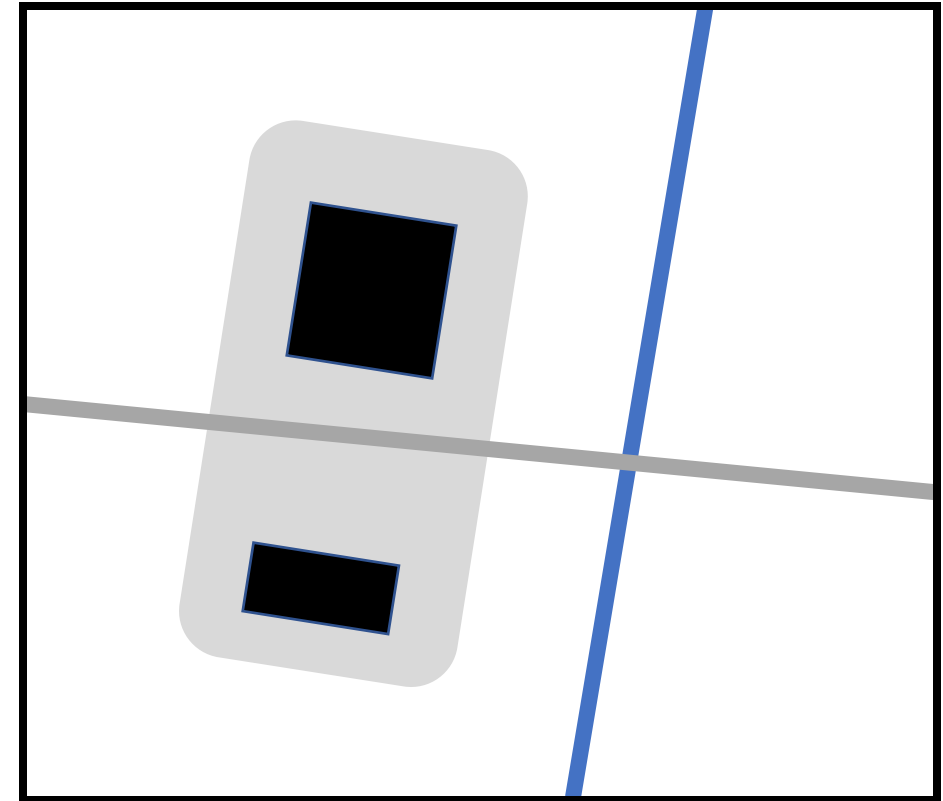
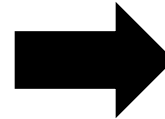


Carte pan-scalaire

Généralisation cartographique

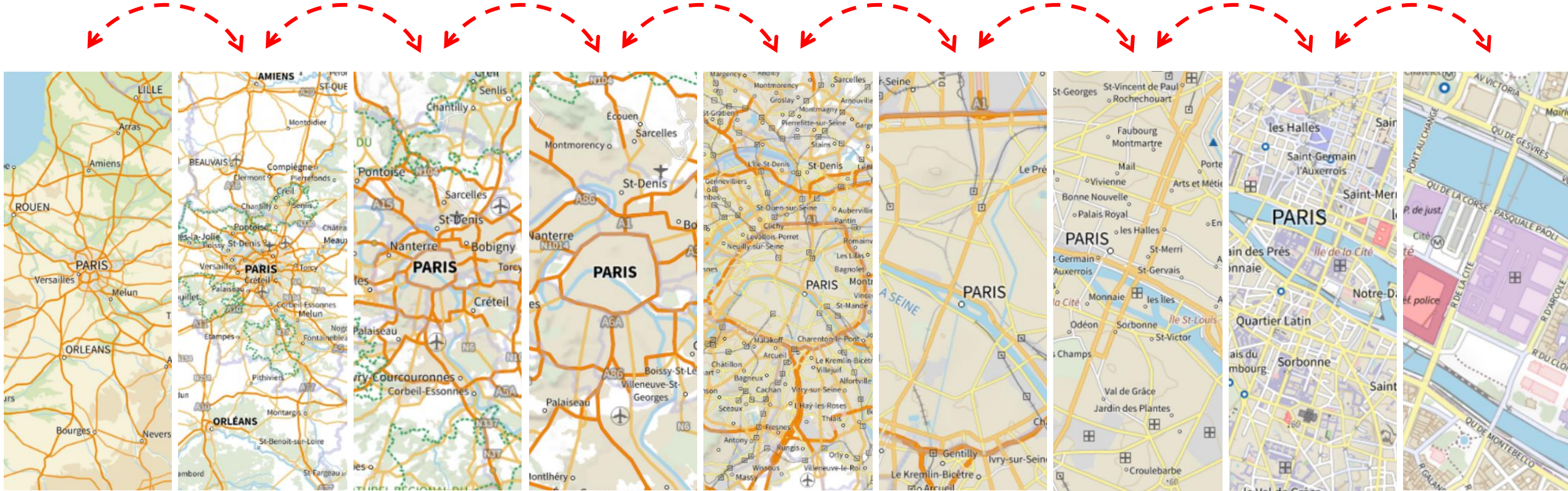


Niveau détaillé



Niveau visé

Progressivité



Carte pan-scalaire

I) Les **pilliers** de la progressivité

II) Construire une carte **vraiment** progressive et **vraiment pas** progressive

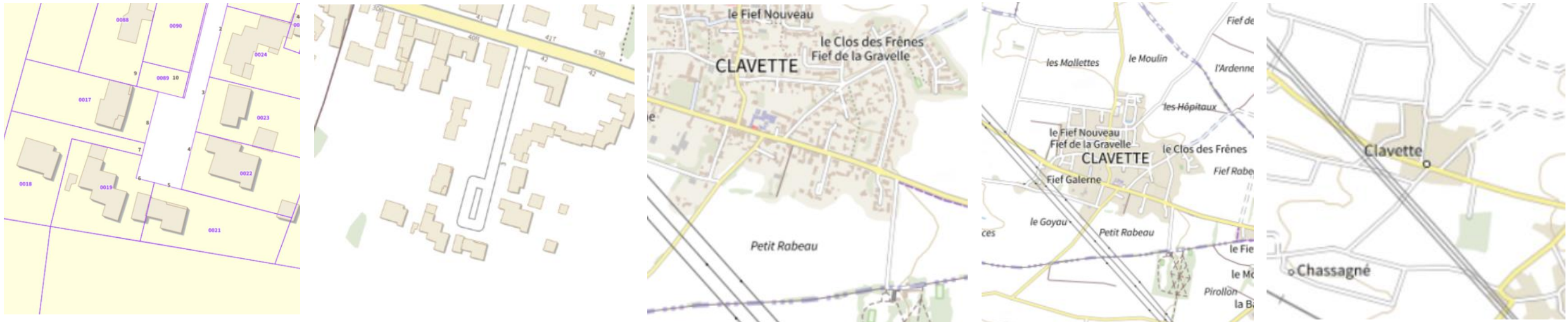
III) Expérimenter la progressivité auprès des **utilisateurs** et **designers**

I) Les piliers de la progressivité

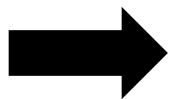


A) La séquence

« The Shining », de Stanley Kubrick en 1980



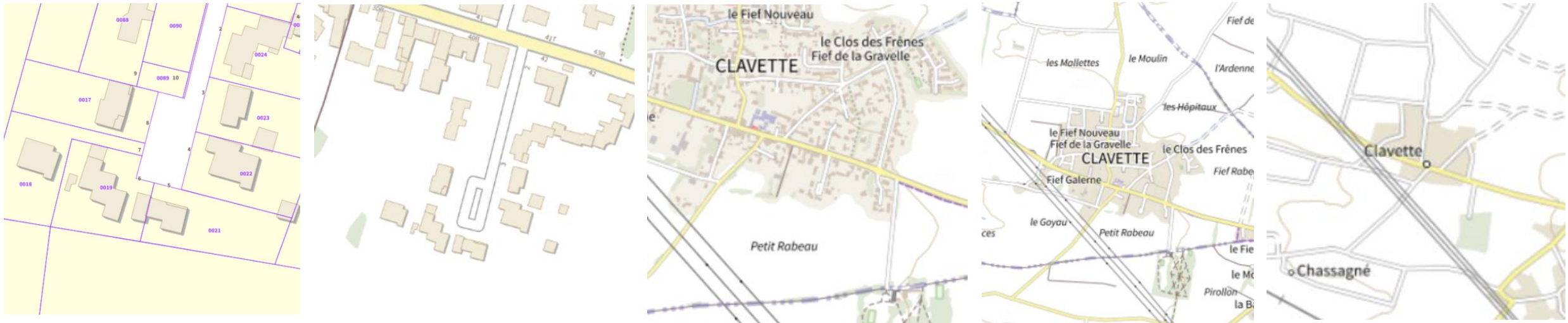
Plan IGN



intervalles d'échelles dans lesquelles un type de données dispose d'une représentation cohérente à travers les échelles

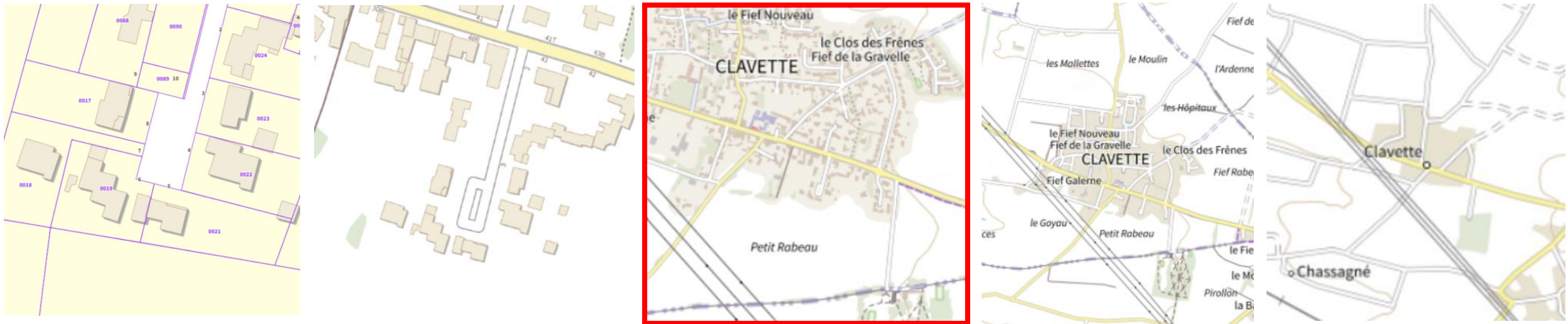
A) La séquence

« The Shining », de Stanley Kubrick en 1980



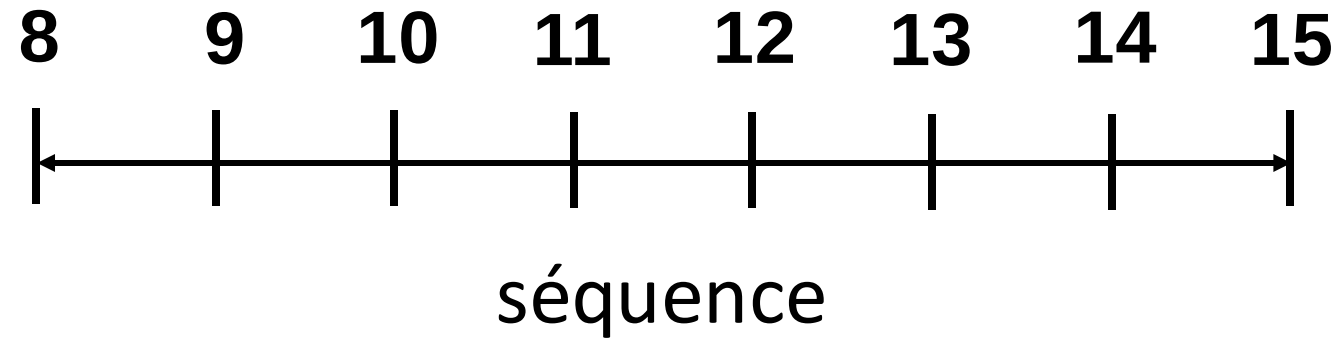
A) La séquence

« The Shining », de Stanley Kubrick en 1980

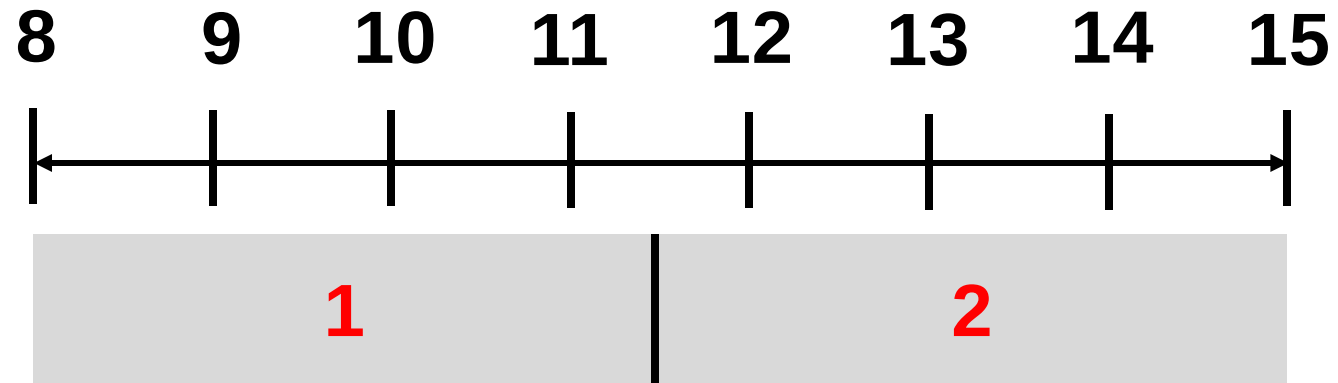


césure

B) Le **rythme**

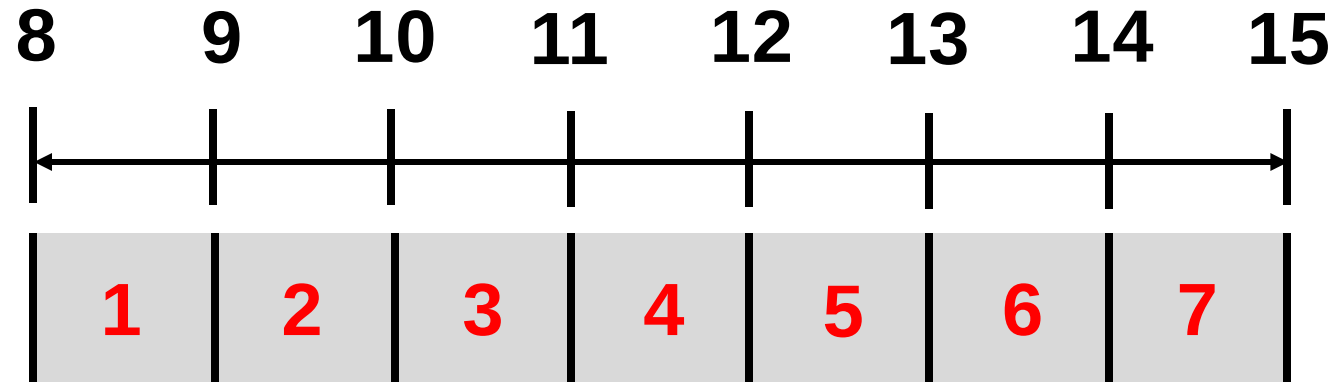


B) Le **rythme**



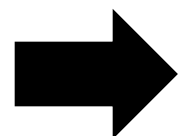
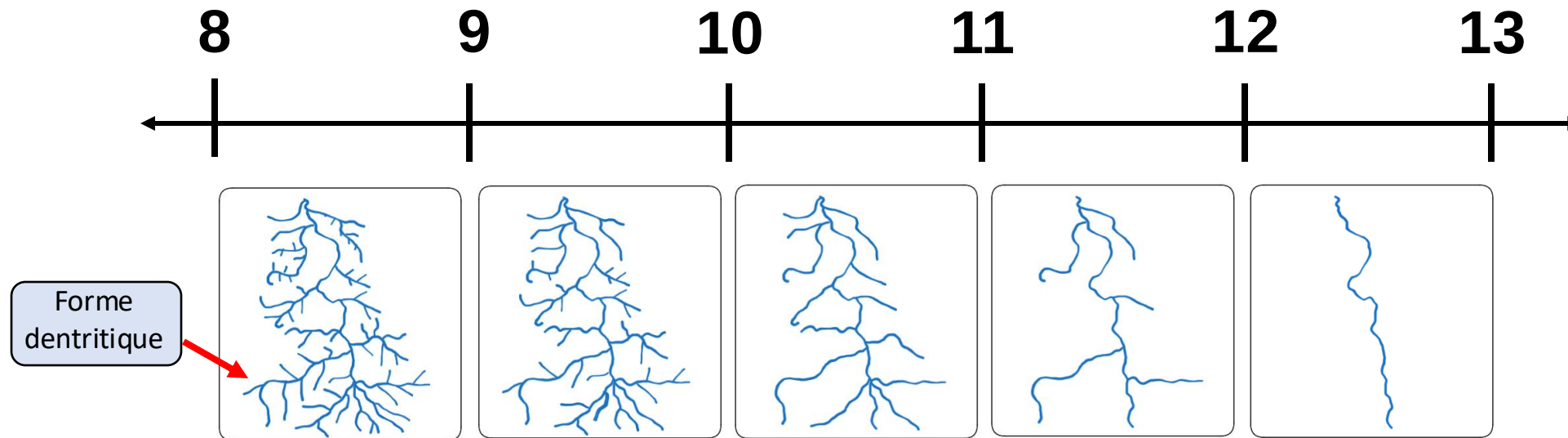
➔ **Fréquence** de changement **lente**

B) Le **rythme**



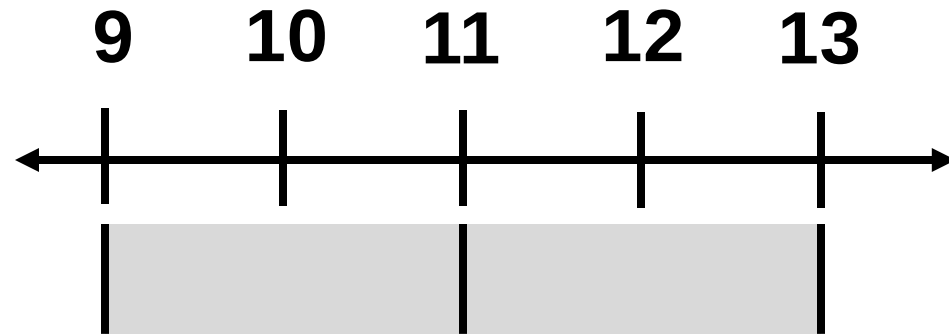
➡ **Fréquence** de changement **rapide**

C) Le sens



Maintien morphologique

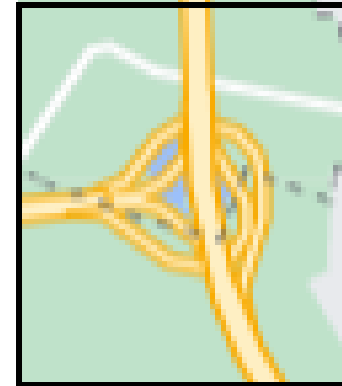
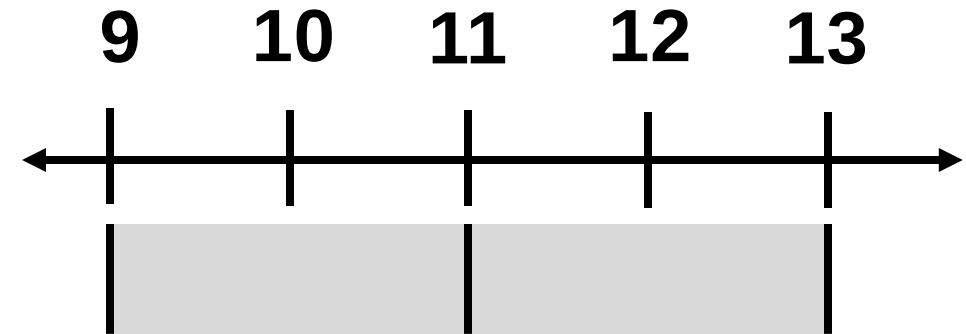
D) La **cohérence**



Geoportail Luxembourg



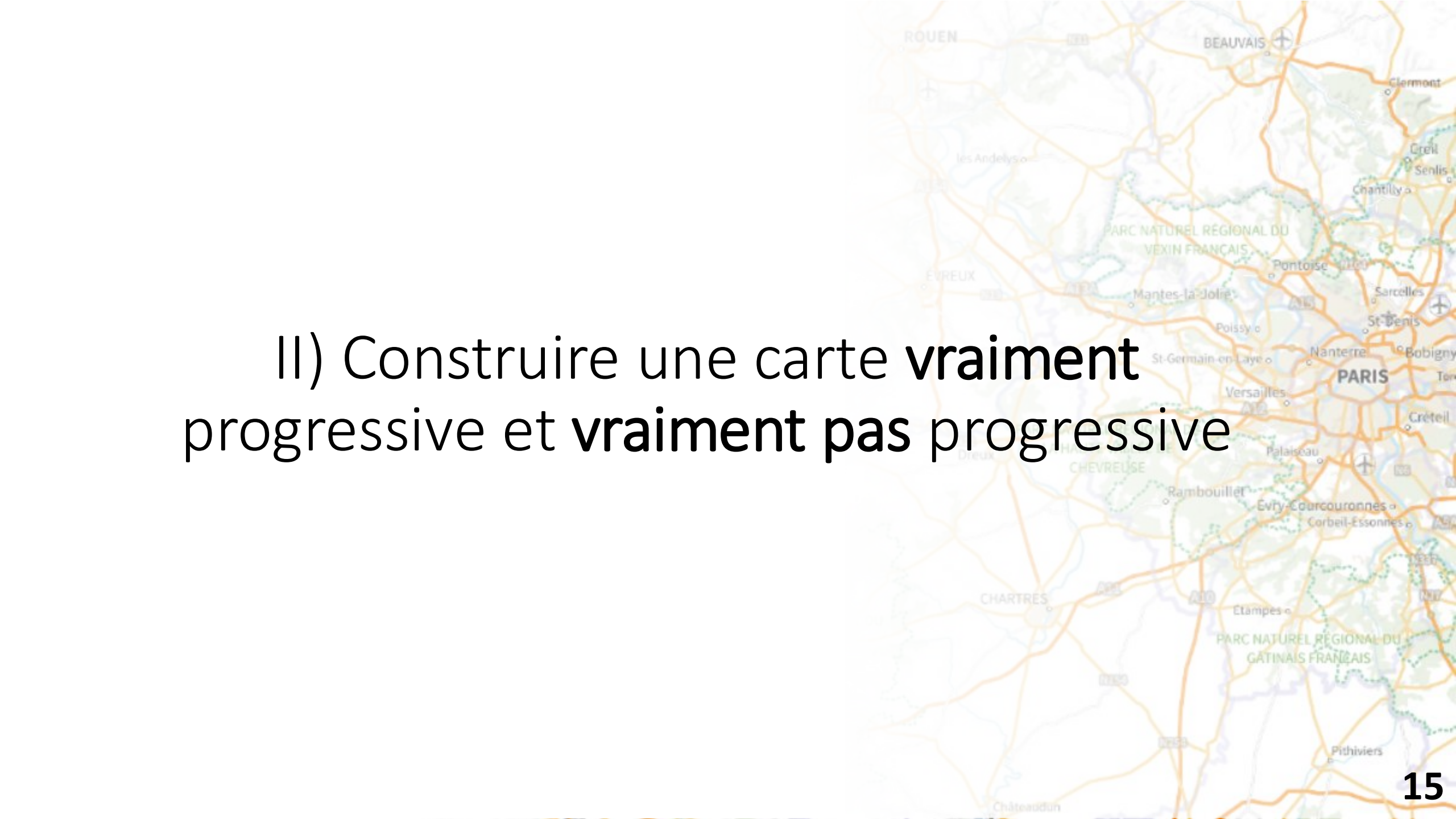
➡ Incohérence de style



Google Maps

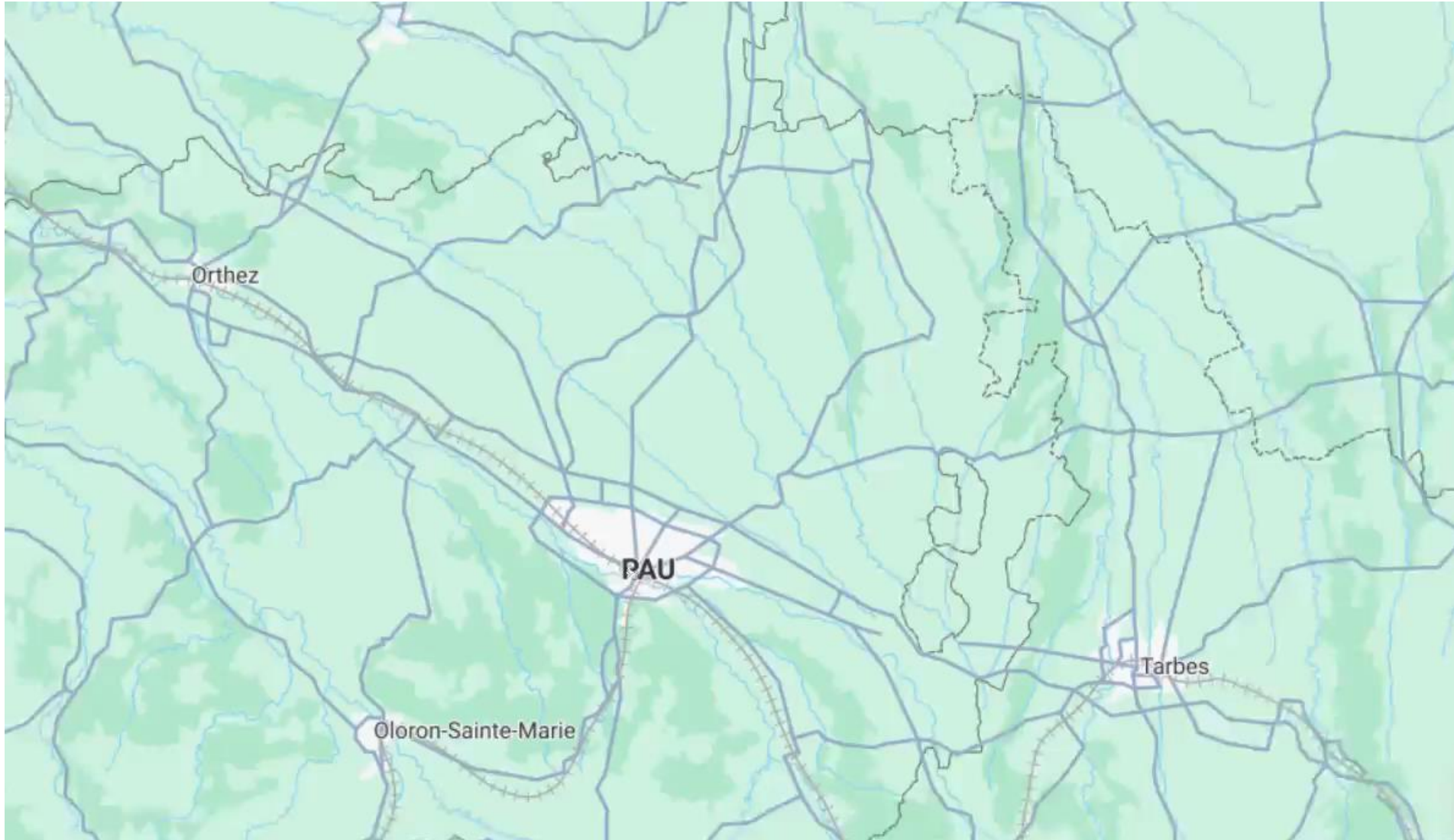
➡ Route continue ? Sans issue ?

La **progressivité** est la minimization des ruptures de **sens**, de **coherence** et de **rythme**



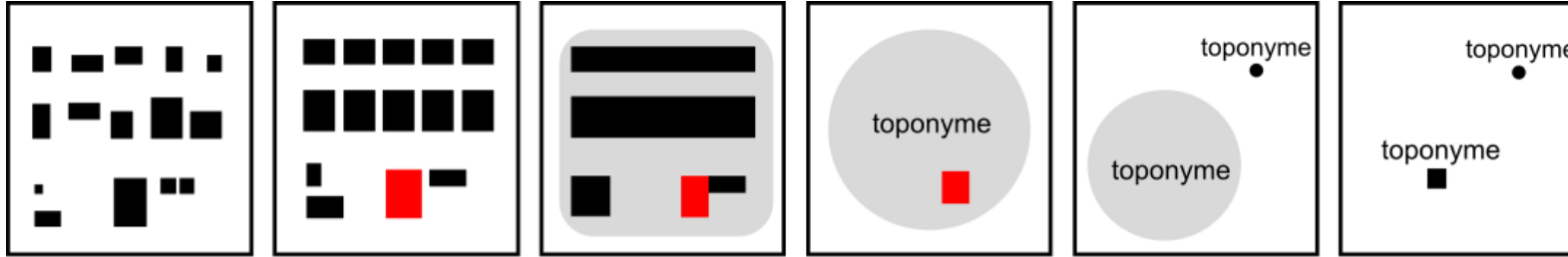
II) Construire une carte **vraiment**
progressive et **vraiment pas** progressive

Une carte **vraiment** progressive : la bonne carte

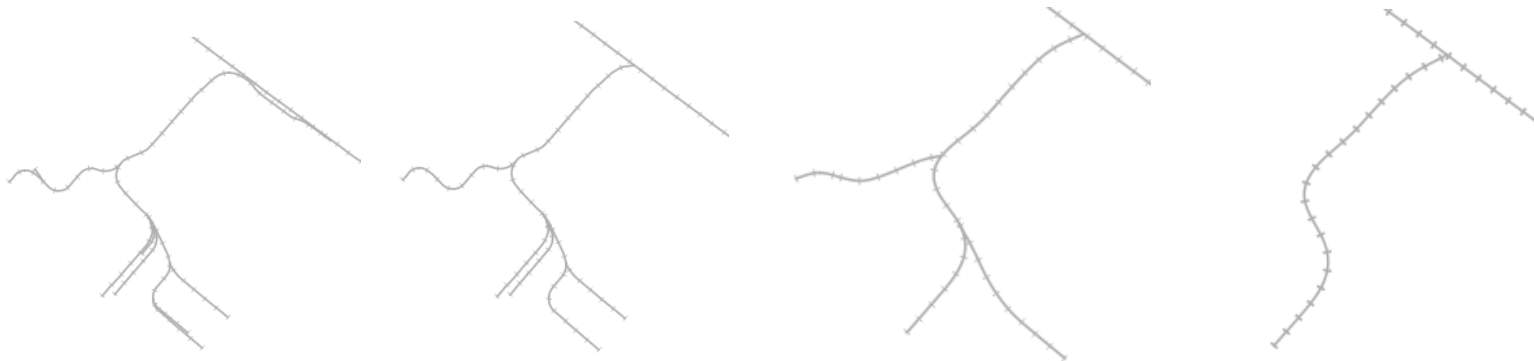


Sources : BD TOPO + Algorithmes de Cartagen

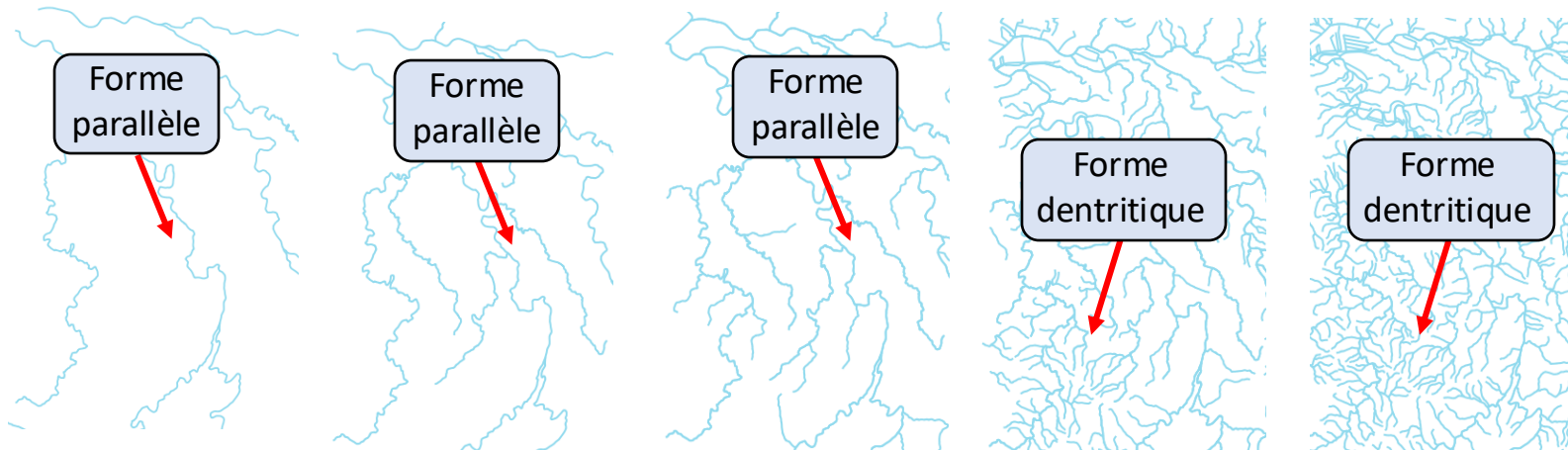
Une carte **vraiment** progressive



Travailler en
séquences

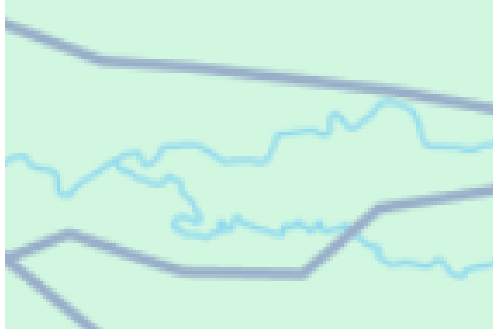


Fréquence **rapide** +
rythme constant

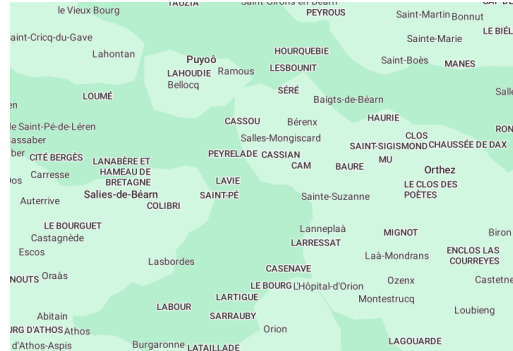
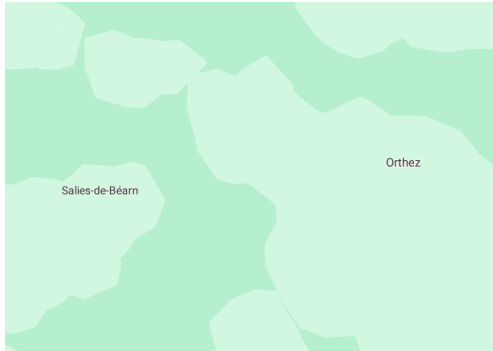


Préservation
morphologique

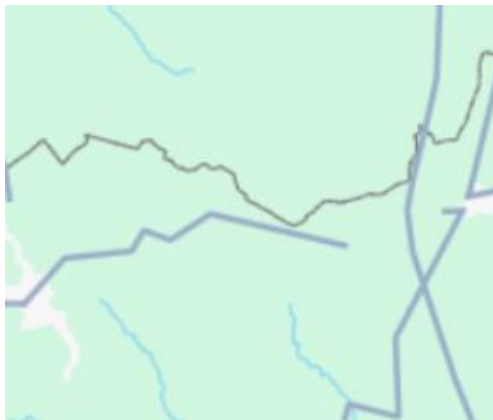
Une carte **vraiment pas** progressive : Les **RUPTURES**



→ déconnexions



→ Rupture rythmique



→ Incohérence stylistique



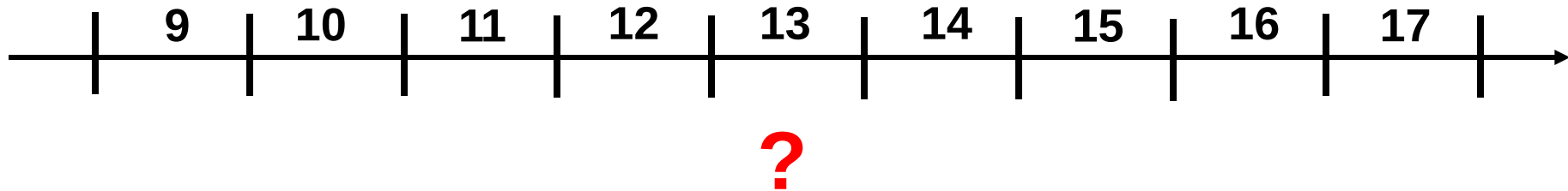
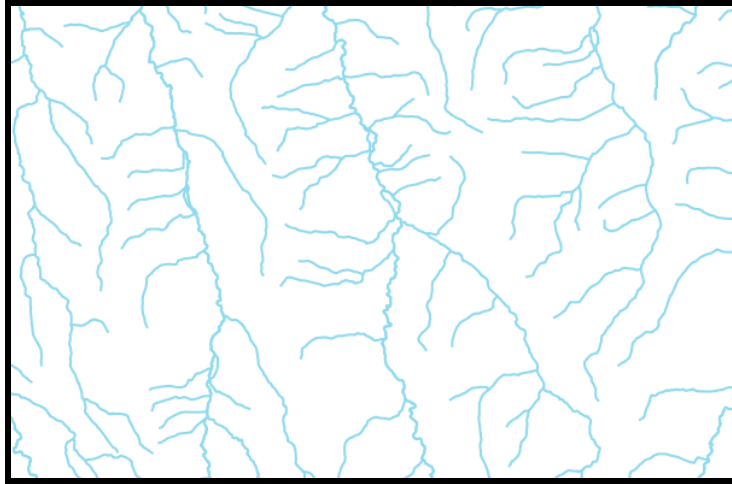
III) Expérimenter la progressivité auprès des utilisateurs et designers

Expérimenter la progressivité auprès des **designers**

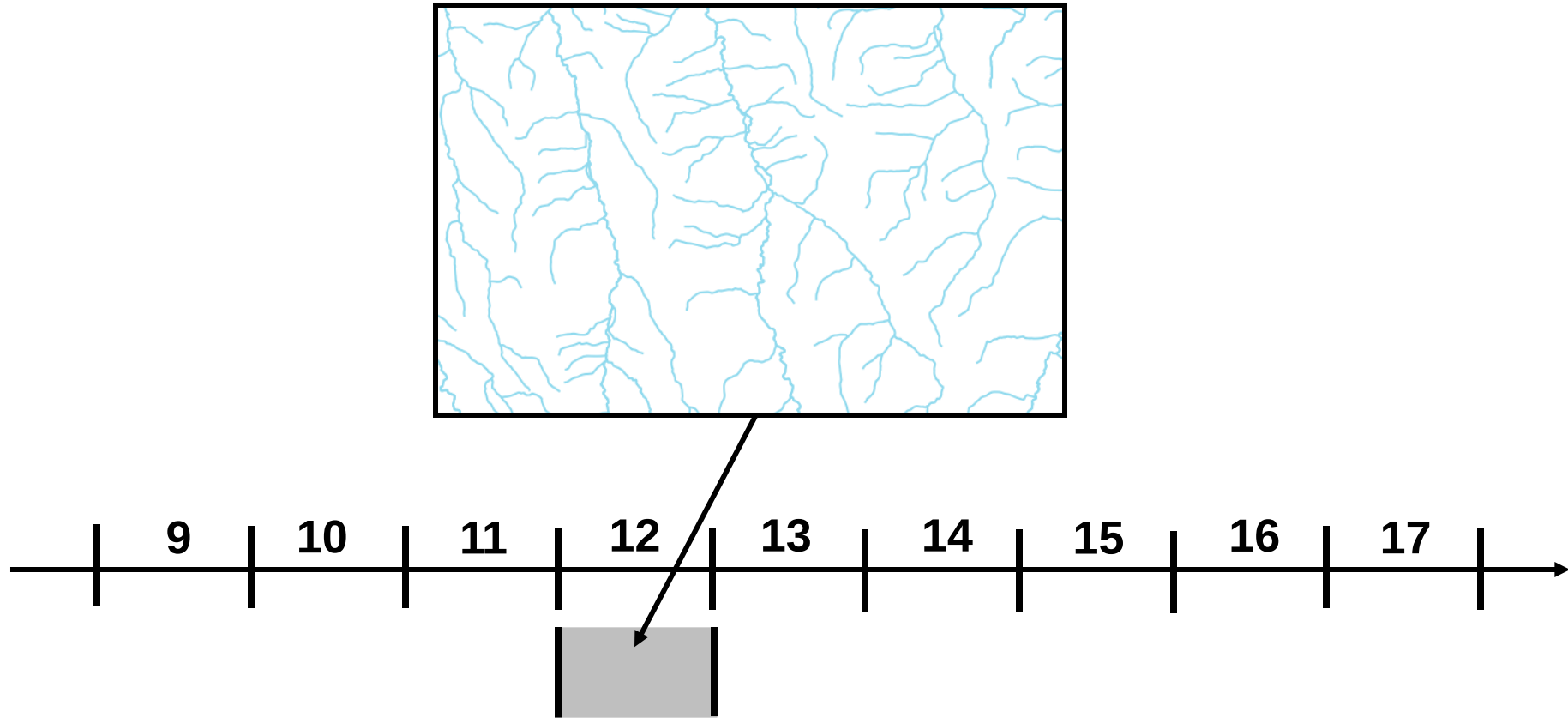


La bonne carte

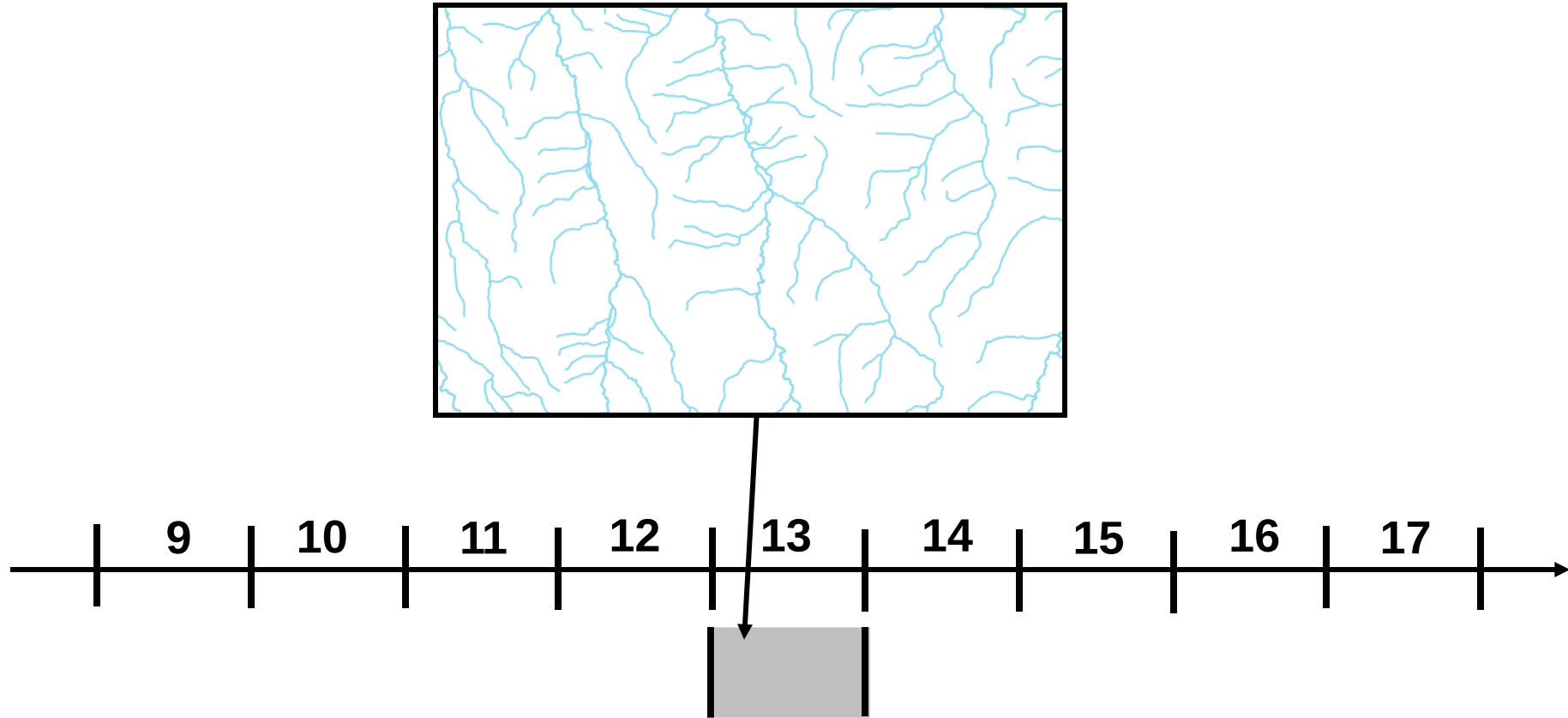
Quelle echelle pour quelle représentation ?



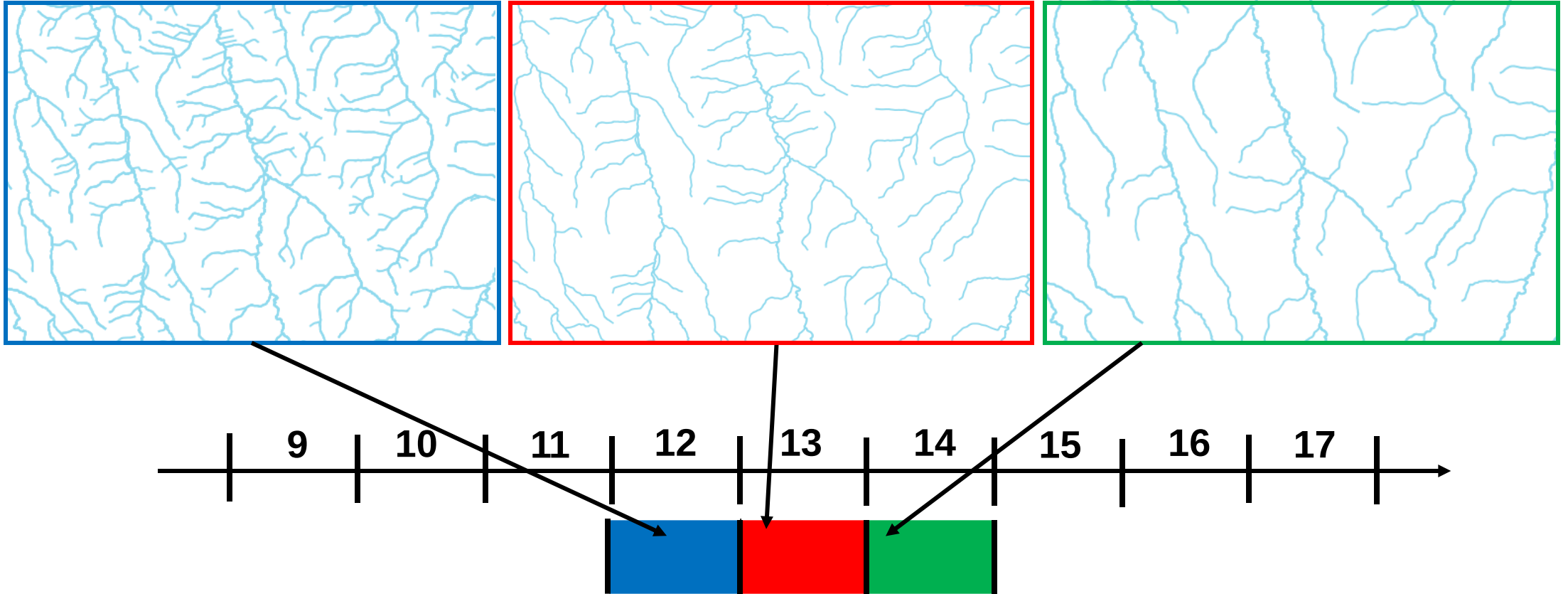
Quelle echelle pour quelle représentation ?



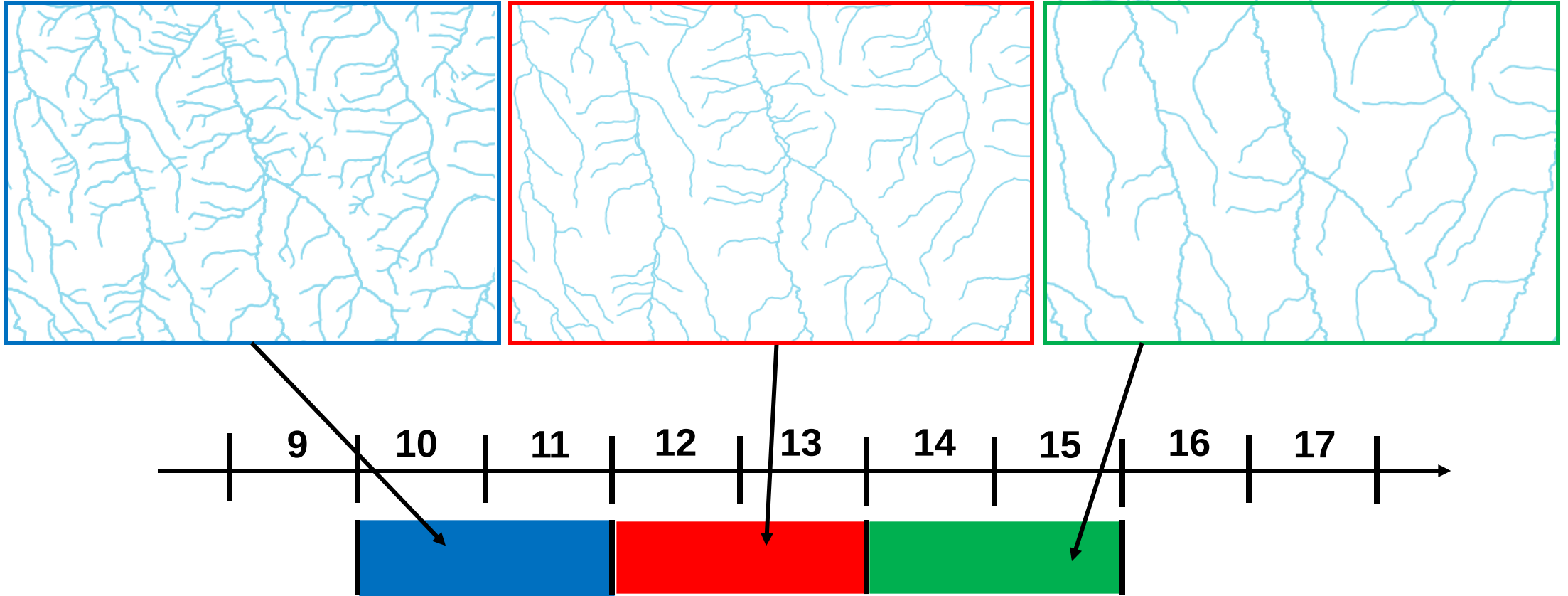
Expérimenter la progressivité auprès des **designers**



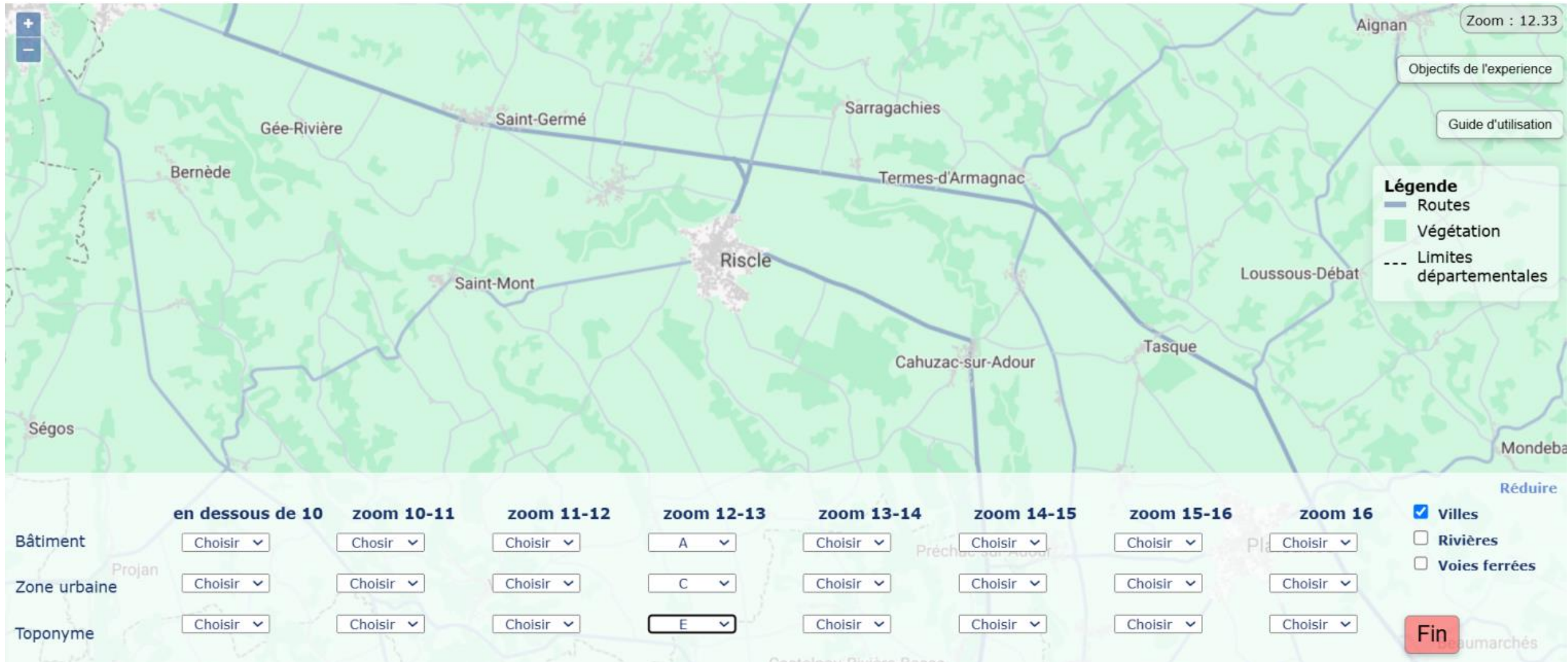
Quel rythme pour chaque représentation?



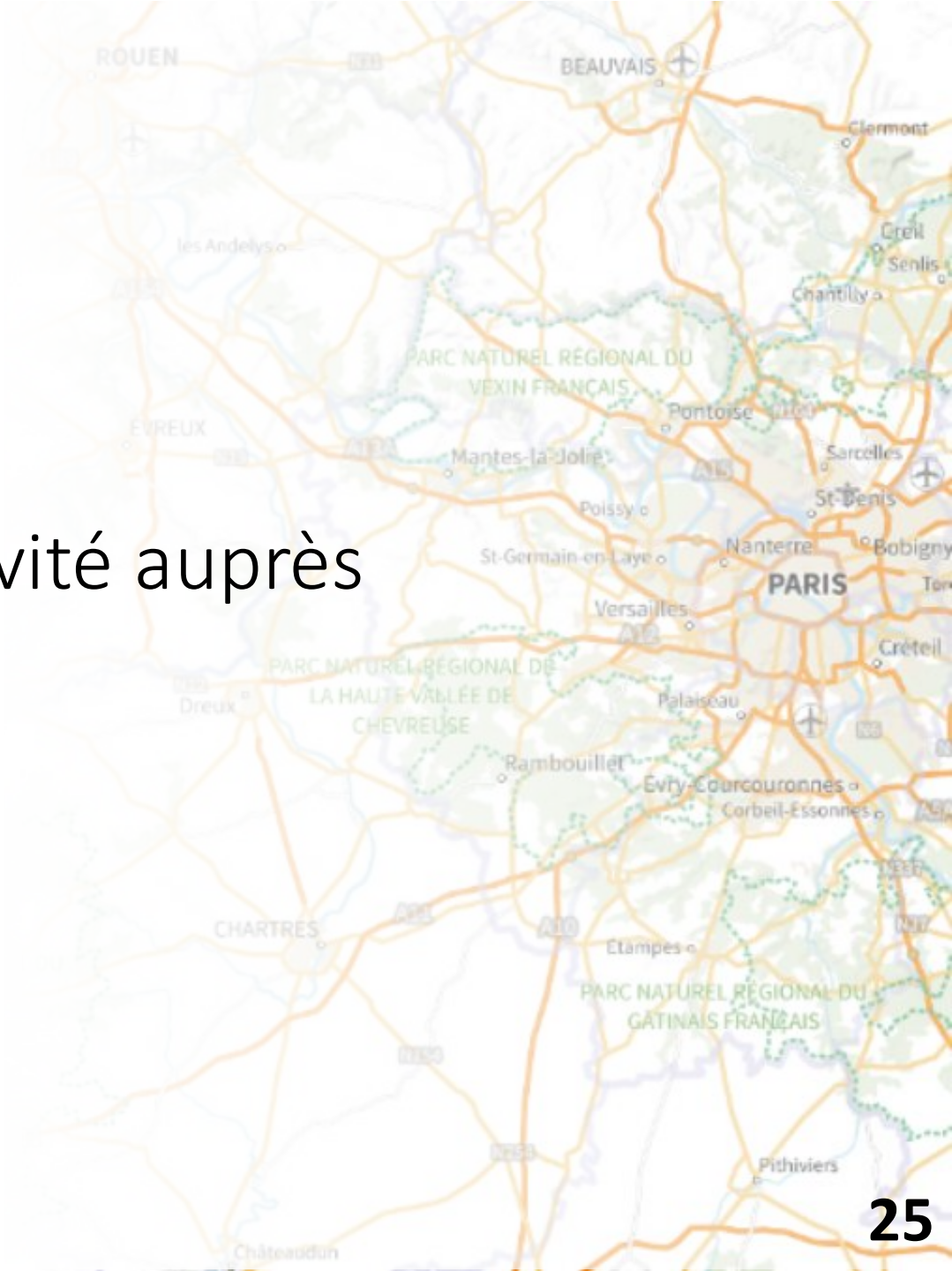
Quel rythme pour chaque représentation?



Expérimenter la progressivité auprès des designers



Expérimenter la progressivité auprès des **utilisateurs**

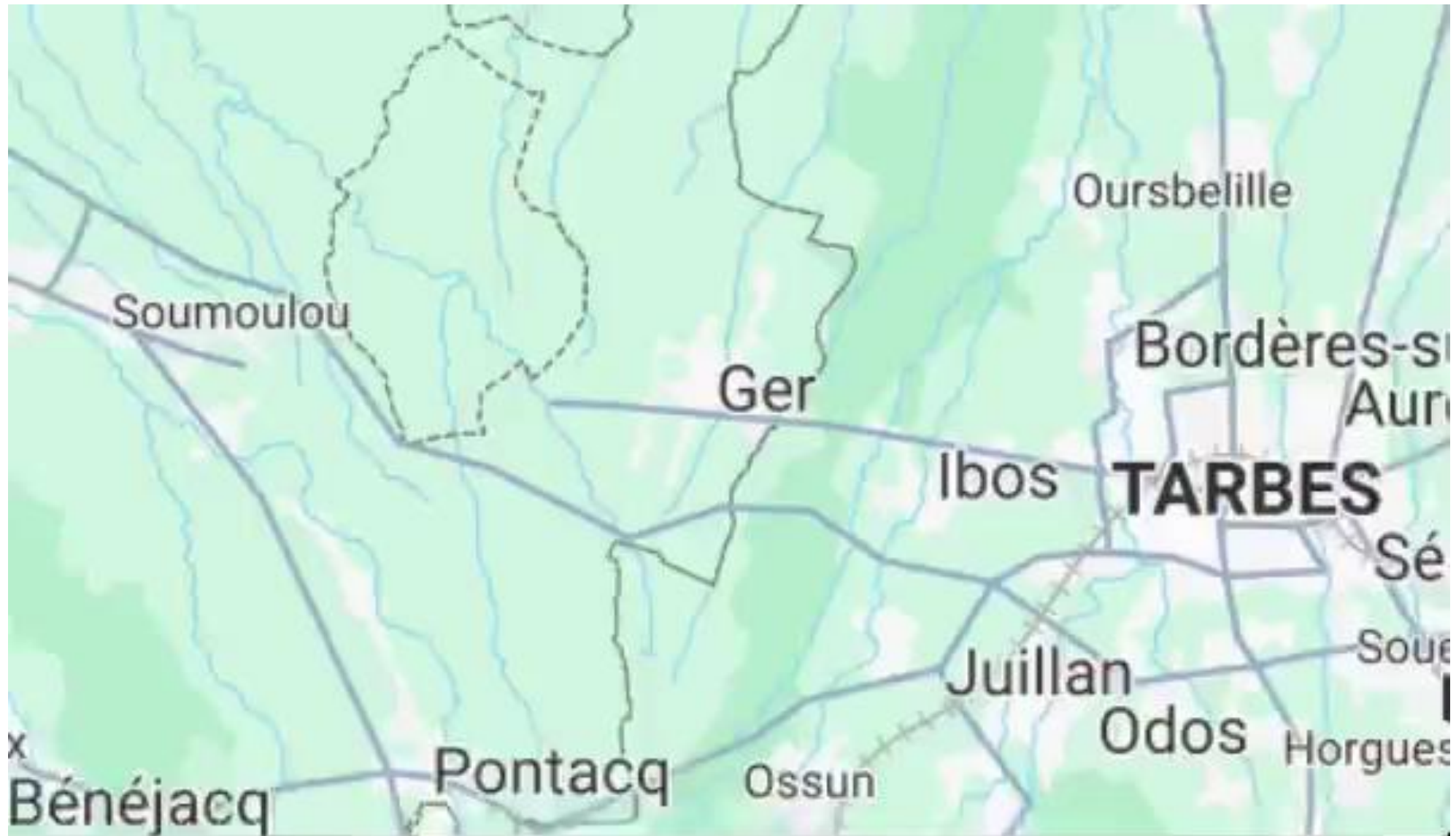


Cherchez le faux-raccord

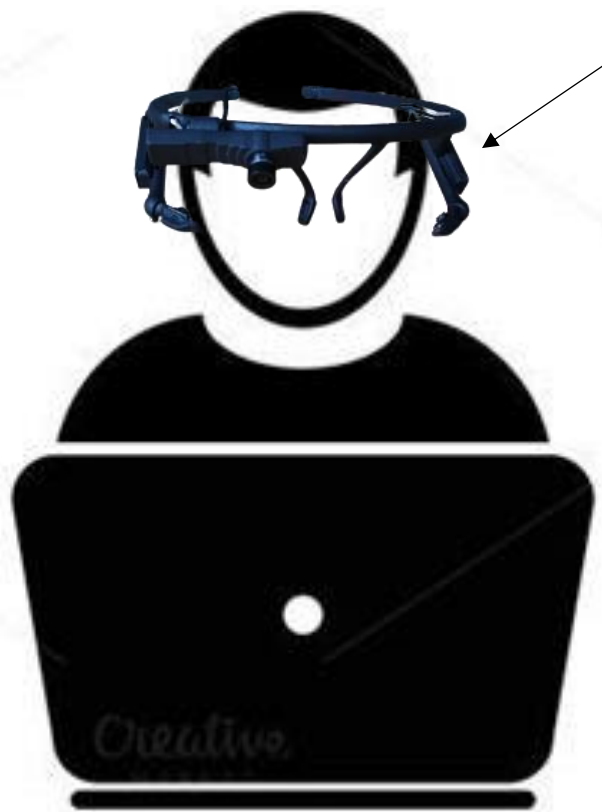


Pinocchio, 1940

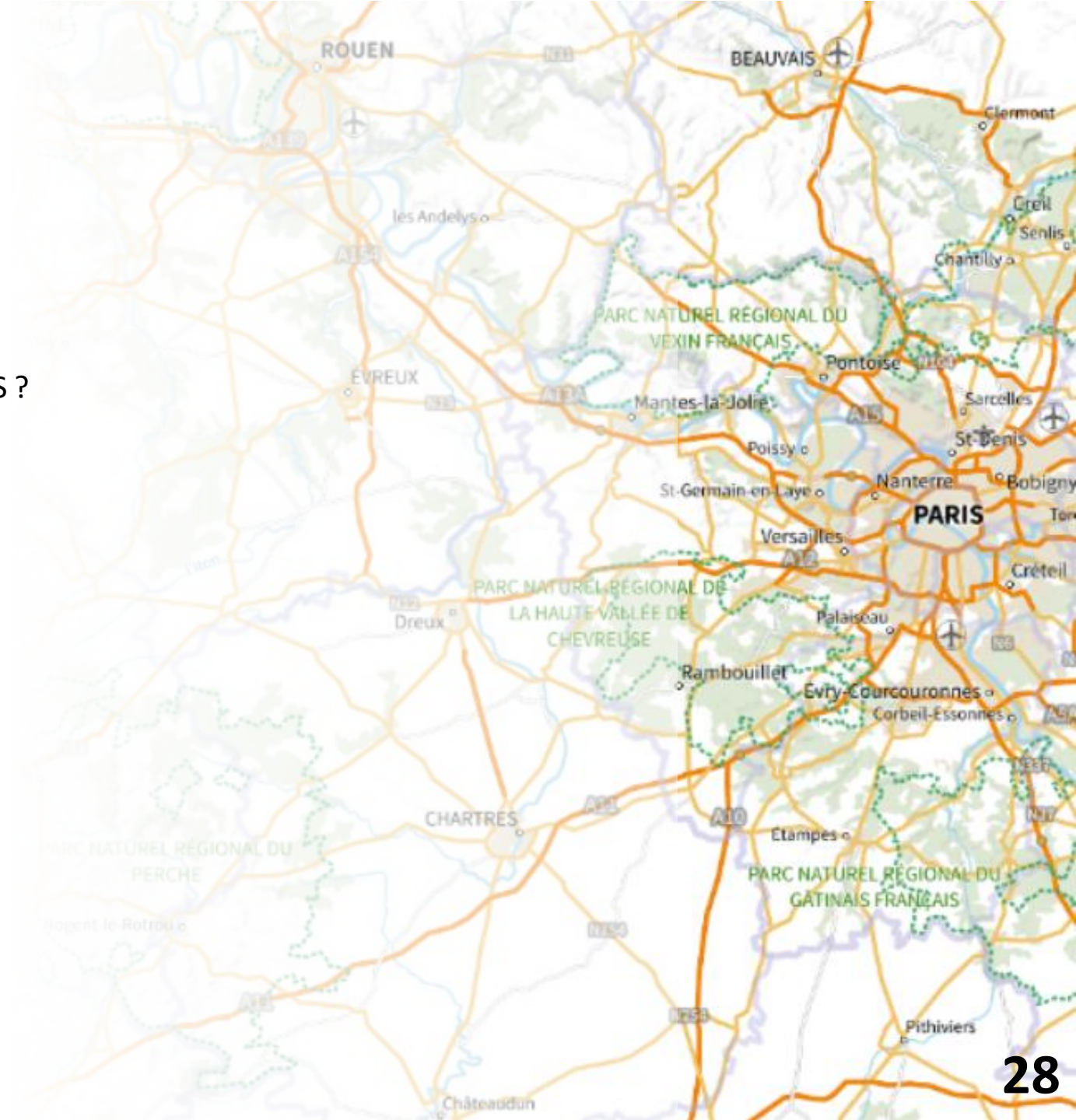
Cherchez les RUPTURES



CONCLUSION



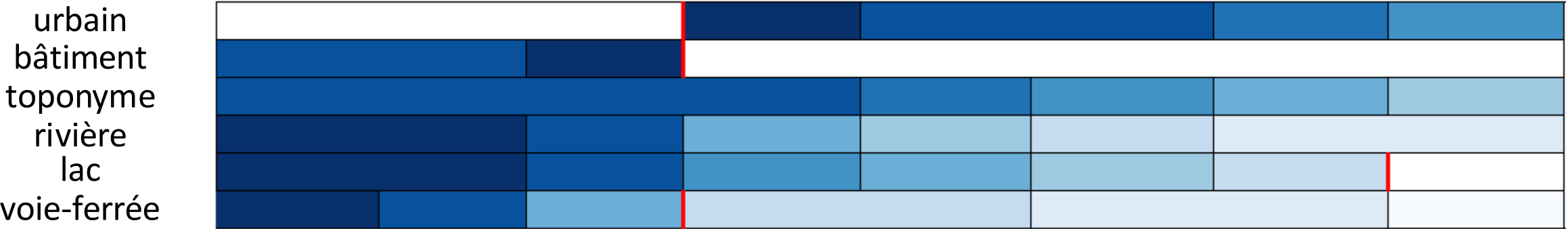
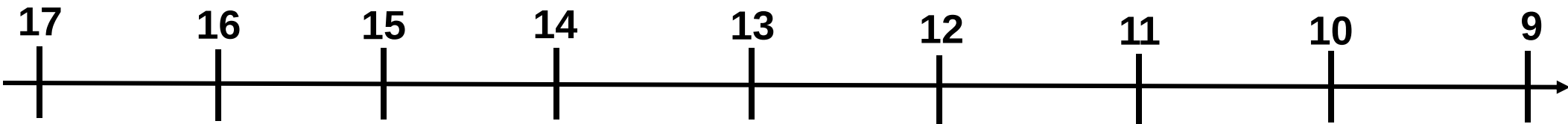
VOUS ?



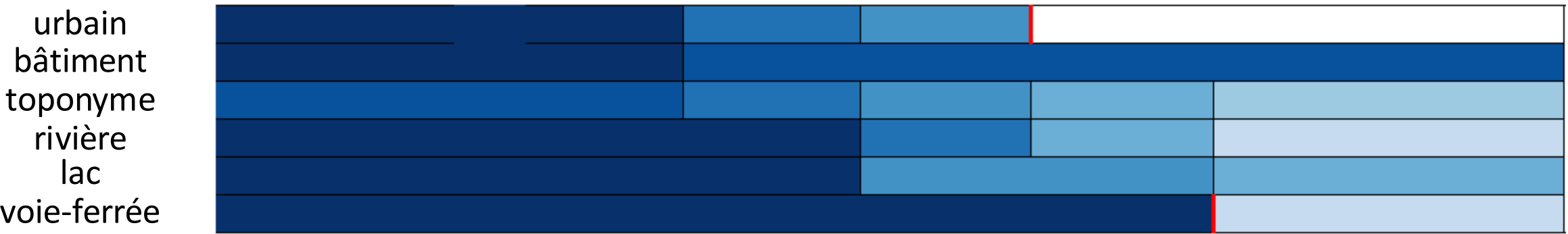
Merci de votre
attention



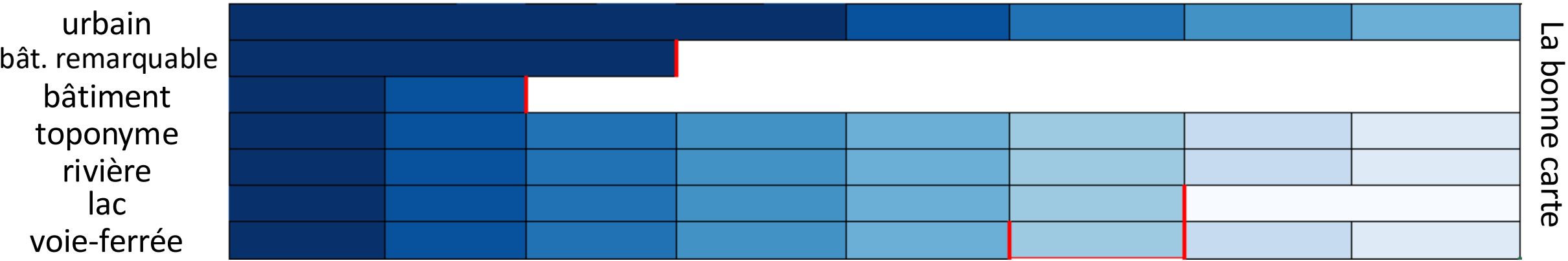
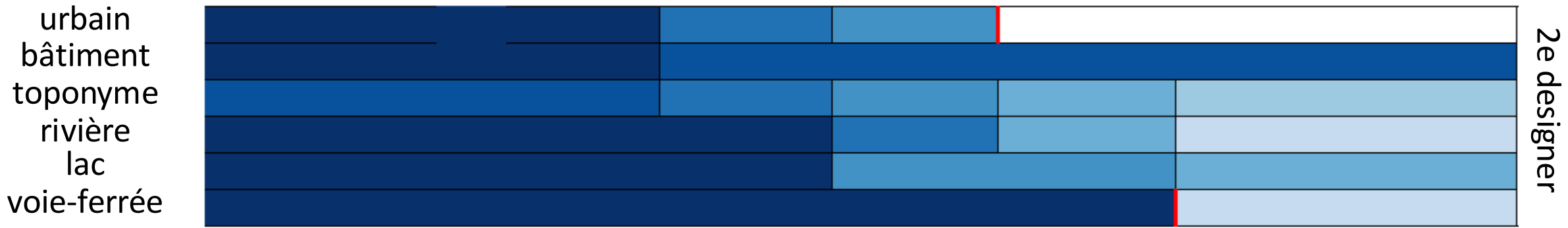
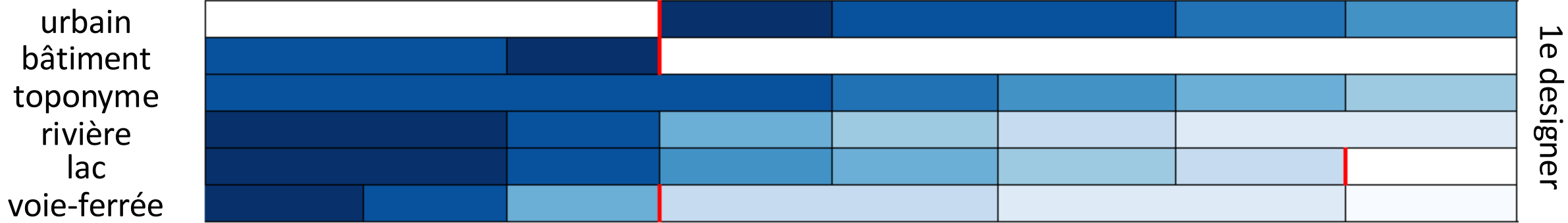
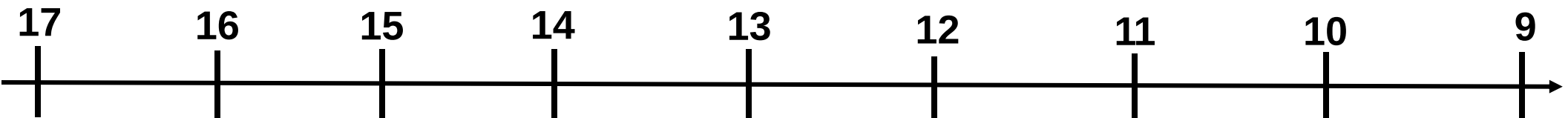
Premiers résultats - ScaleMaster

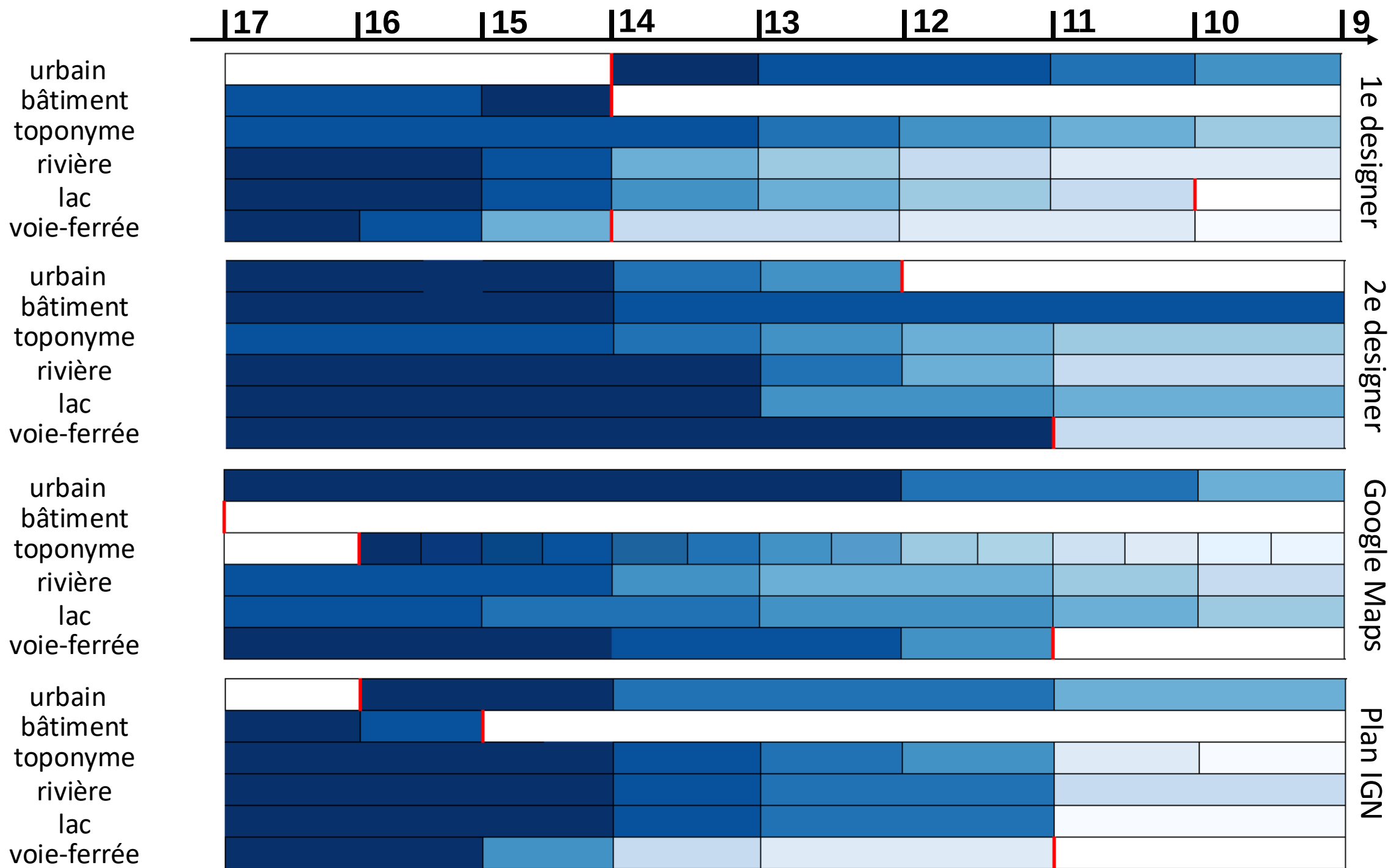


1e designer

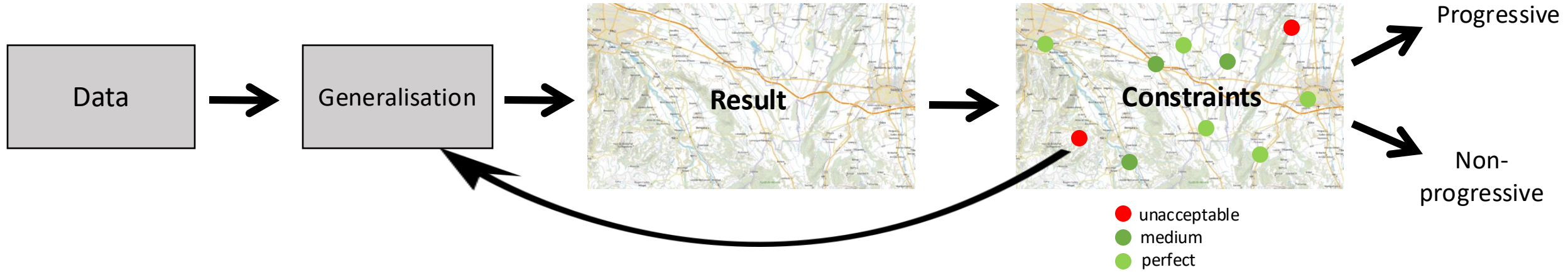


2e designer



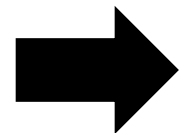
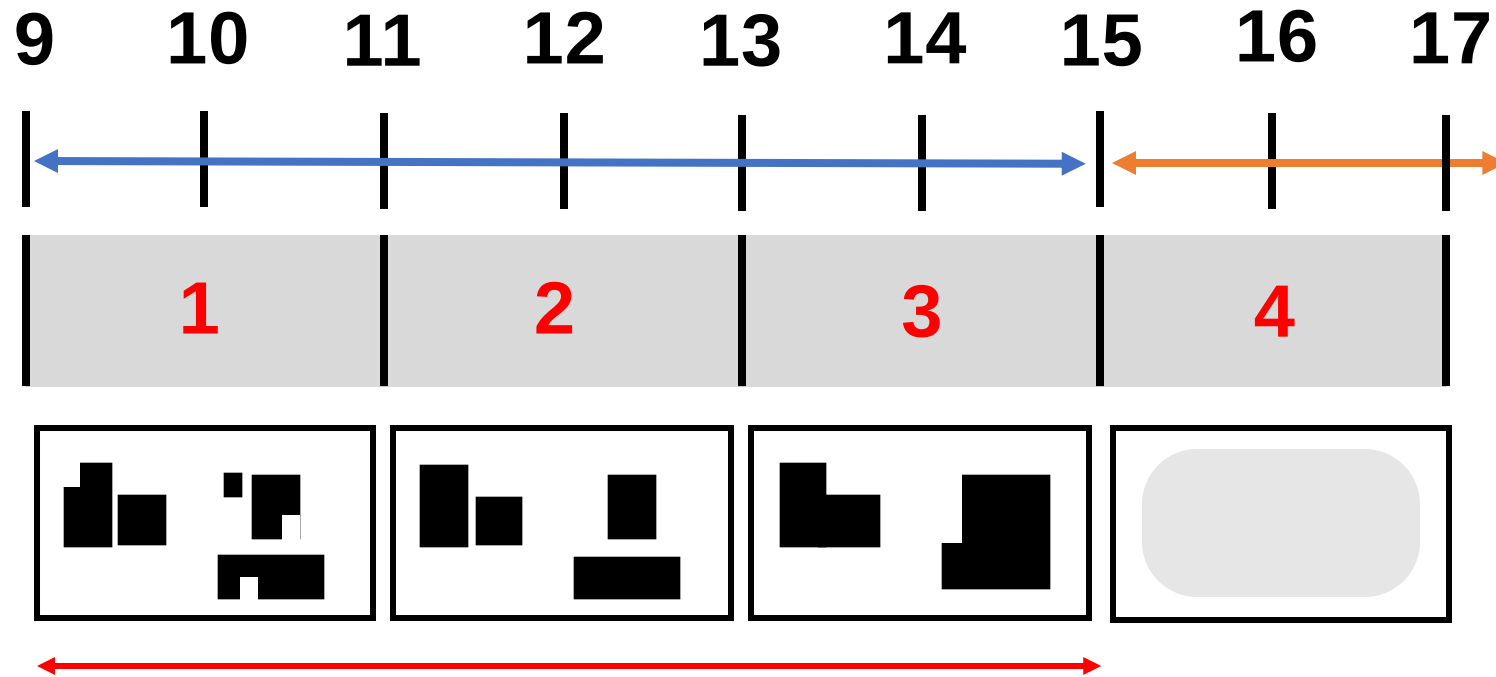


Evaluate progressiveness with automatic constraints



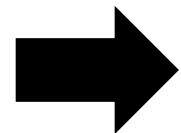
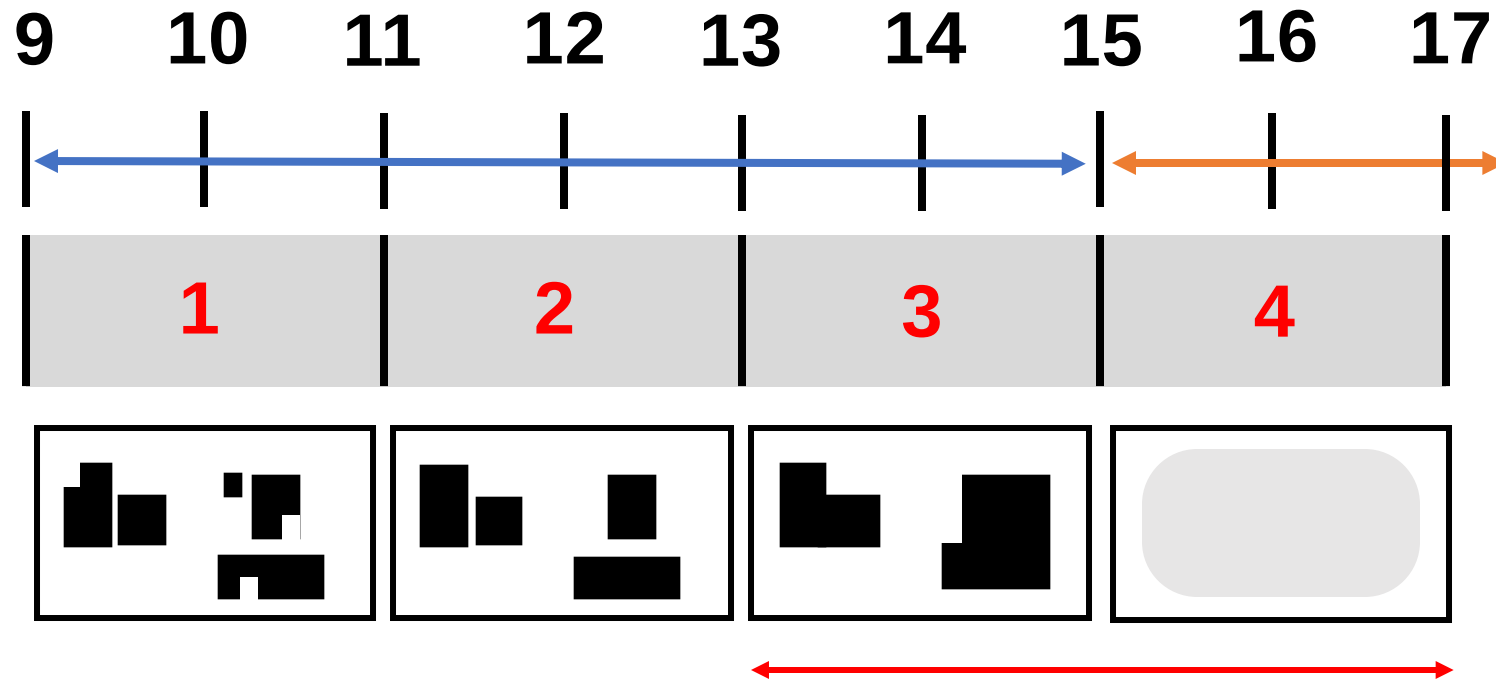
- Density constraint
- Pattern constraint

B) Le **rythme**



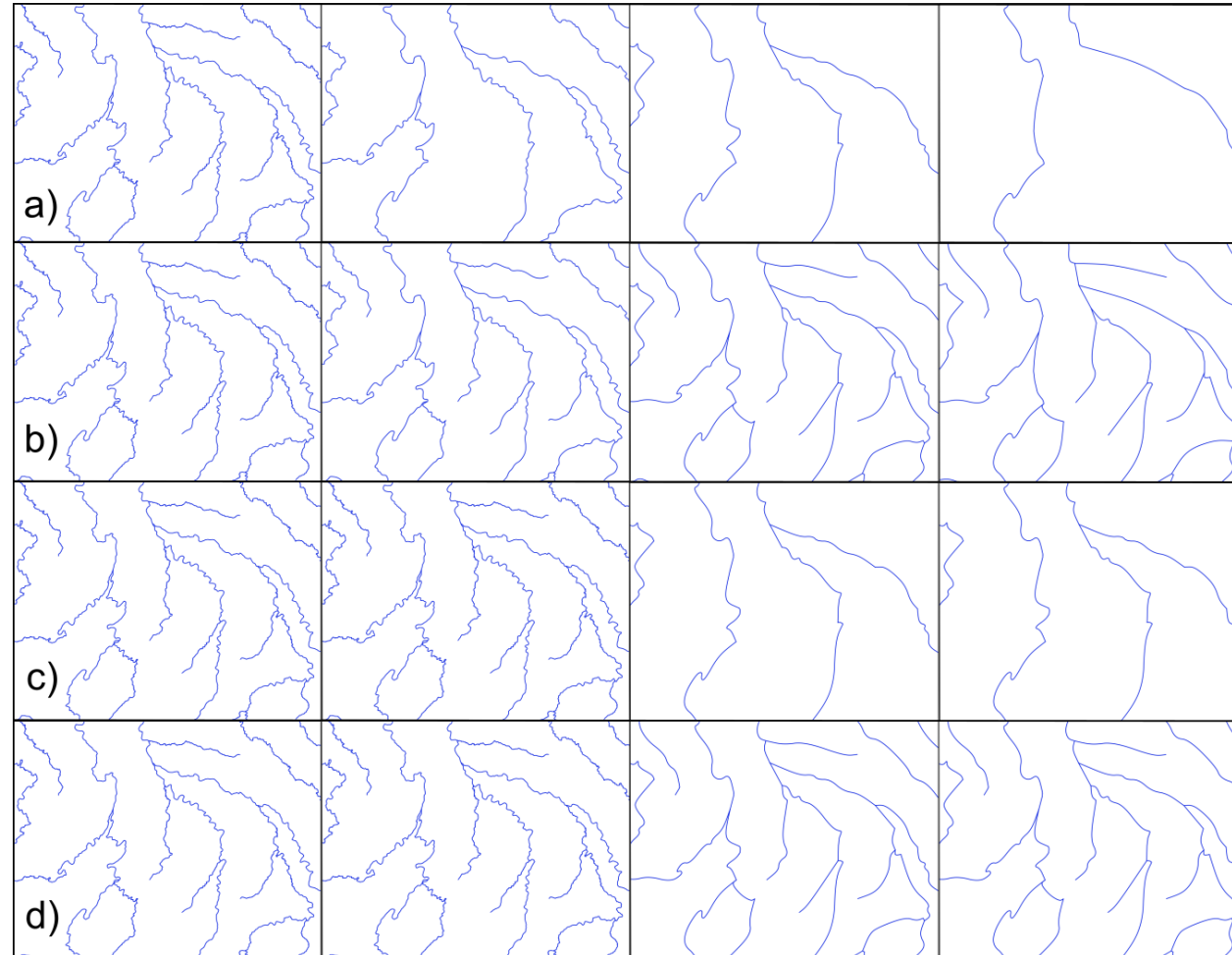
Faible **amplitude** de changement

B) Le **rythme**



Forte **amplitude** de changement

Evaluate progressiveness : A user experiment on different maps



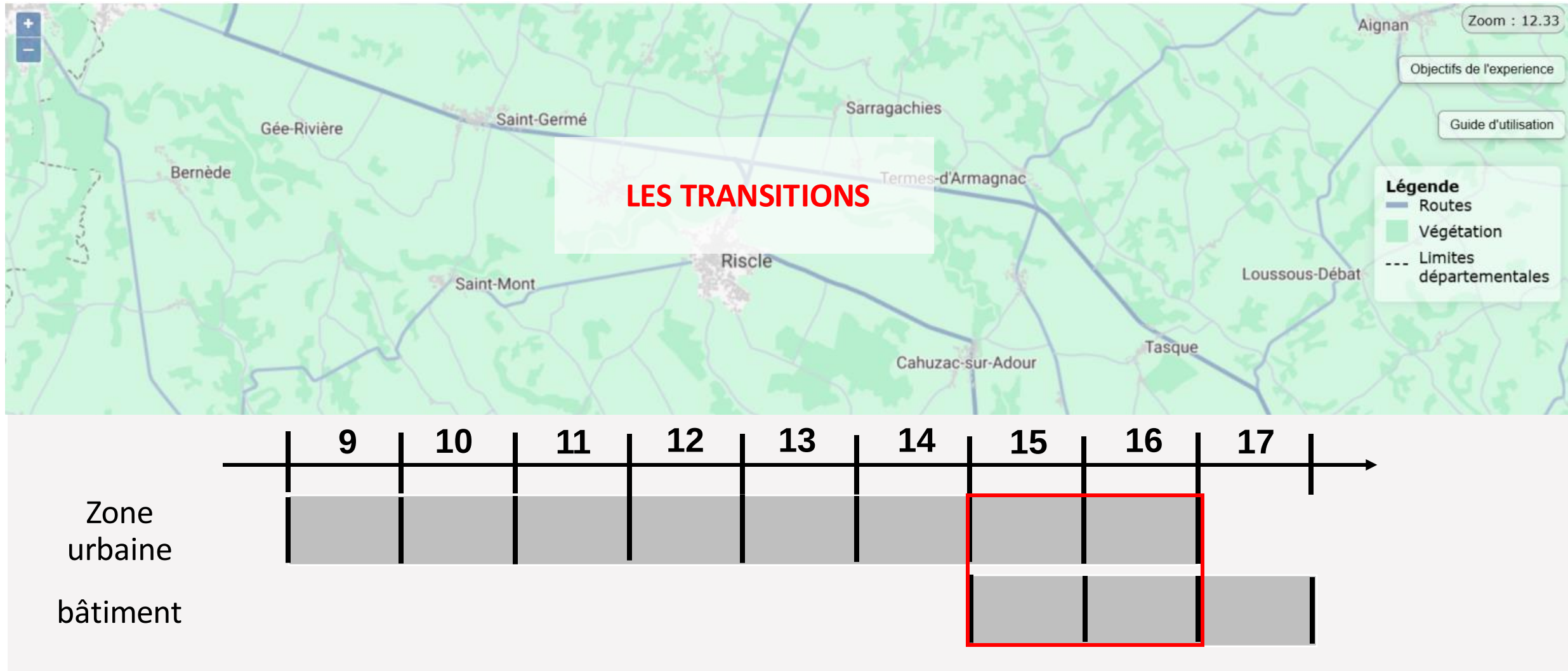
fast map cadence and fast abstraction cadence

fast map cadence and slow abstraction cadence

slow map cadence and fast abstraction cadence

slow map cadence and slow abstraction cadence

Expérimenter la progressivité auprès des designers



Learn progressiveness from music

Flute

Oboe

Clarinet

Bassoon

The image shows a conductor score for the first movement of the Peer Gynt Suite, Op. 46, by Edvard Grieg. The score is for the woodwind section, specifically the Flute (Fl.), Oboe (Ob.), Clarinet in A (Cl. (A)), and Bassoon (Fag.). The music is in 3/4 time and D major. The tempo is marked 'piu tranquillo'. The score is divided into four measures. The first measure features a flute melody starting on a piano (p) dynamic. The second measure introduces the oboe and clarinet with a piano (p) dynamic, followed by a forte piano (fp) dynamic. The third measure continues the woodwind entries with a piano (p) dynamic, followed by a diminuendo (dim.) dynamic. The fourth measure features a bassoon melody starting on a piano (p) dynamic, followed by a piano (p) dynamic. The score includes various musical notations such as notes, rests, and dynamic markings.

Conductor Score of Peer Gynt Suite no. 1, Op. 46 - I. Morning

Learn progressiveness from music

Roads

Rivers

Vegetation

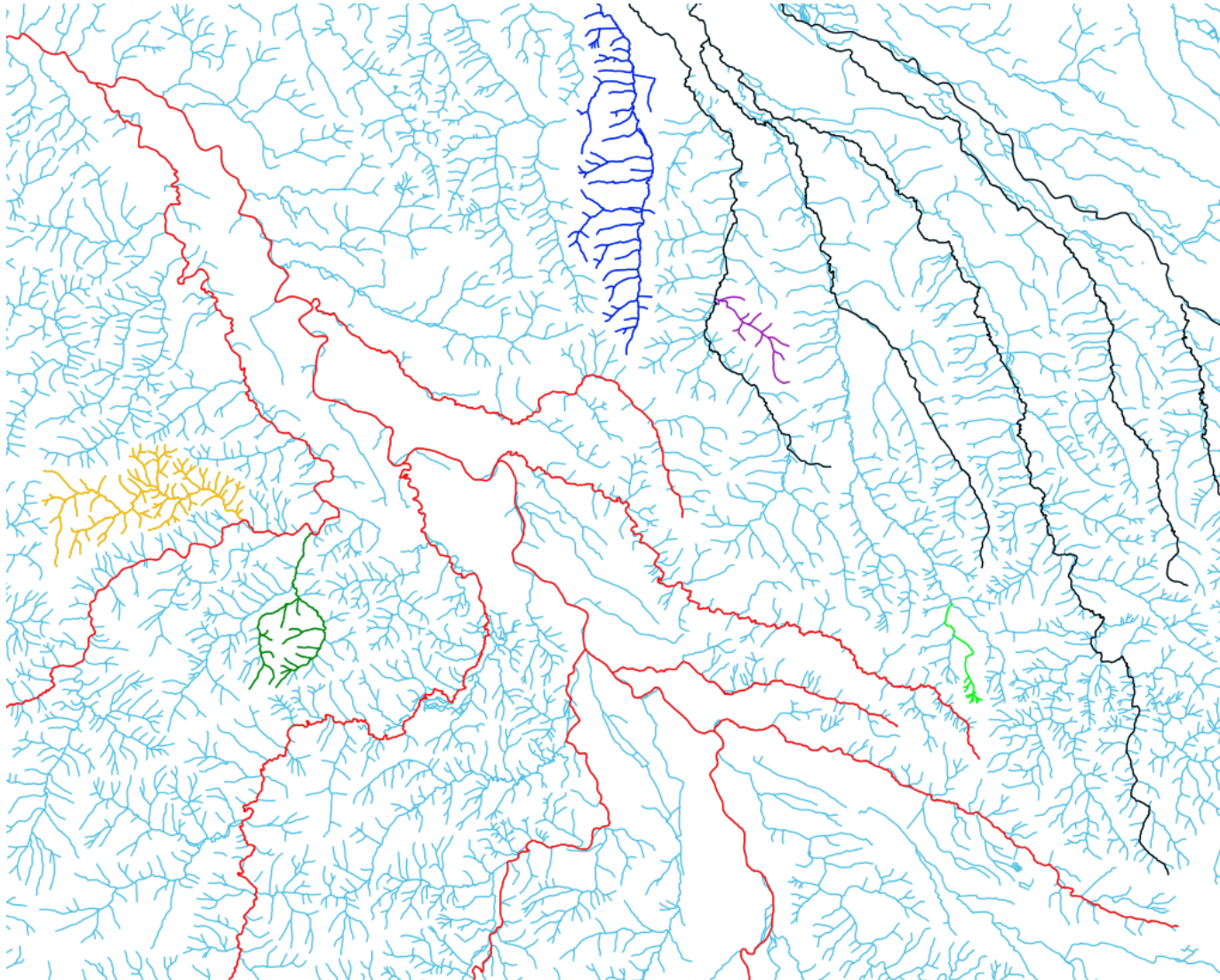
Settlements



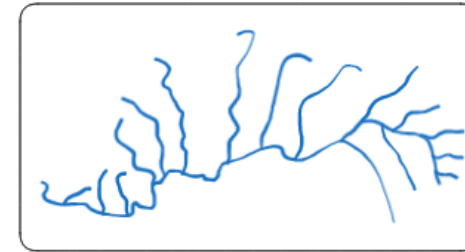
The image displays a conductor score for the first movement of the Peer Gynt Suite, Op. 46, by Edvard Grieg. The score is written for four woodwind parts: Flute (Fl.), Oboe (Ob.), Clarinet in A (Cl. (A)), and Bassoon (Fag.). The key signature is one sharp (F#) and the time signature is 3/4. The tempo/mood is marked 'piu tranquillo'. The score is divided into measures, with dynamic markings such as *p* (piano), *fp* (fortissimo), *dim.* (diminuendo), and *pp* (pianissimo). The Flute part begins with a melodic line marked *p* and *piu tranquillo*. The Oboe, Clarinet, and Bassoon parts enter later, with the Bassoon part marked *p* and *dim.*. The score concludes with a *pp* marking and a *poco rit.* (poco ritardando) instruction.

Conductor Score of Peer Gynt Suite no. 1, Op. 46 - I. Morning

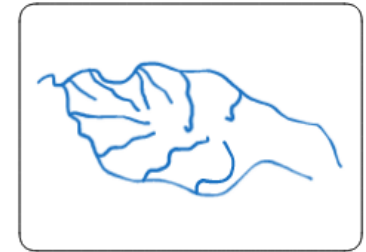
Minimization of breaks of meaning



Our classification of hydrographic patterns in the Adour-Garonne river basin



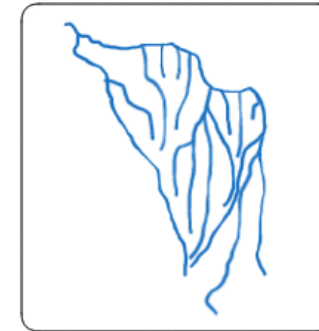
Directional dentritic



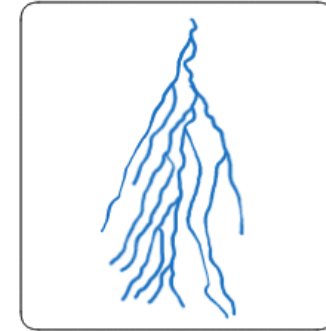
Annular enclave



Trellis



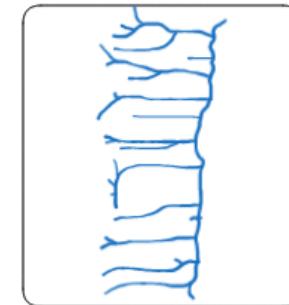
Radial enclave



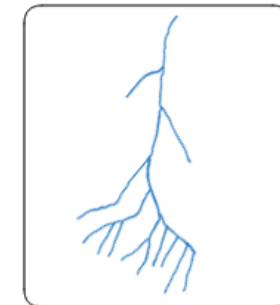
Parallel



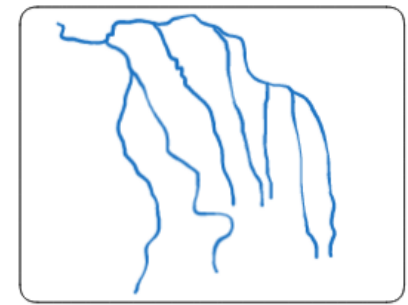
Dentritic



Directional trellis



Accumulated trellis

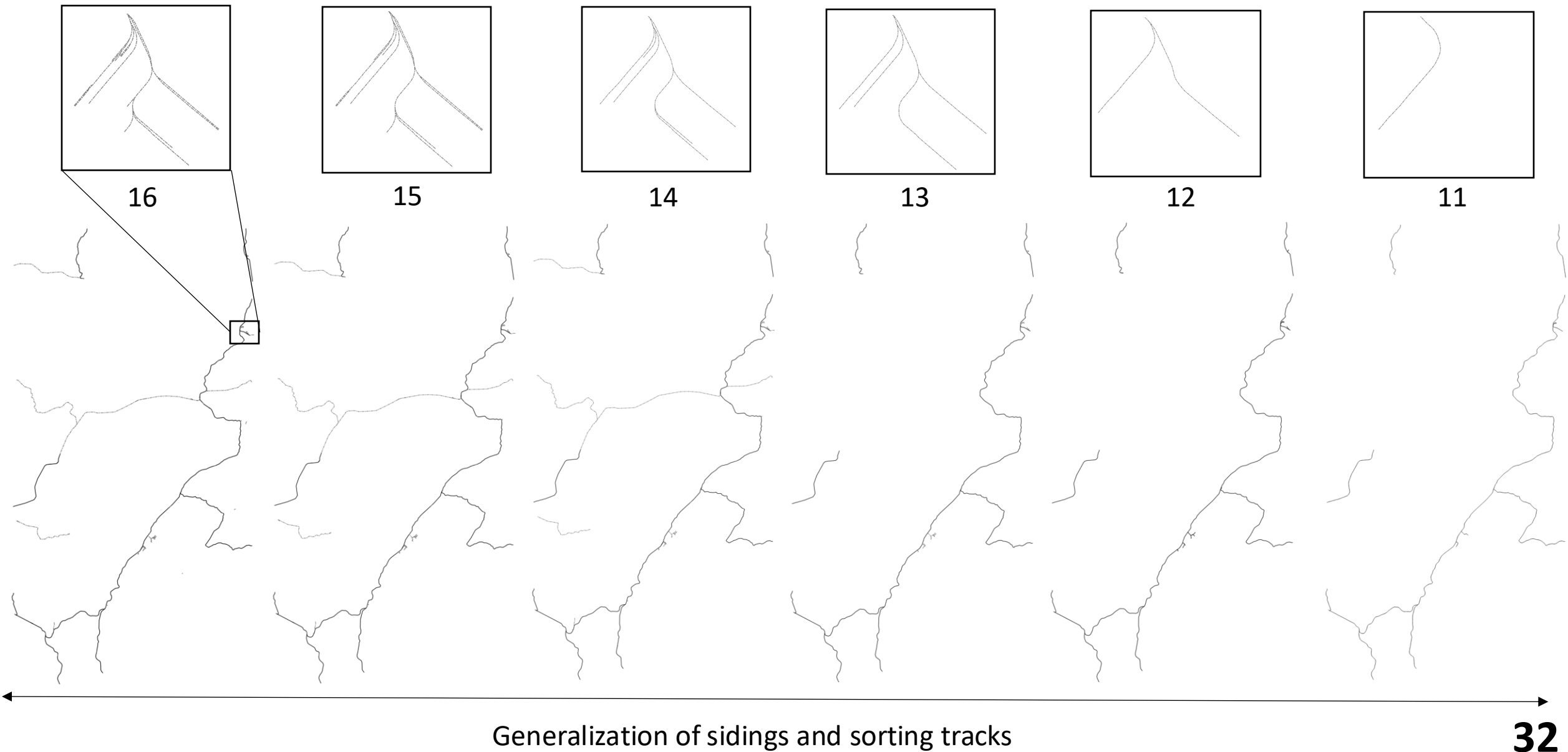


Directional Parallel



Meaning through patterns

Generalization of railways



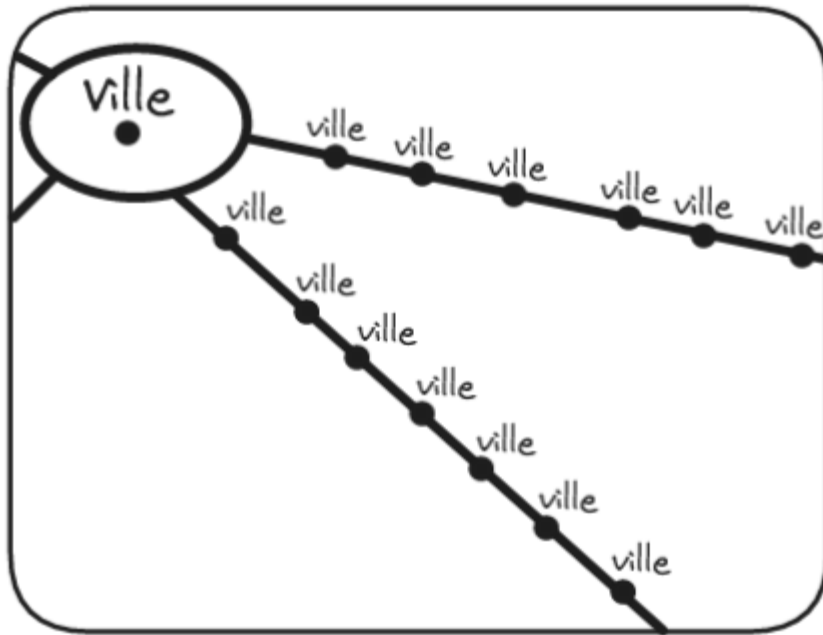
Learn progressiveness from music

VERSE	CHORUS	VERSE	CHORUS	BRIDGE	CHORUS
A	B	A	B	C	B

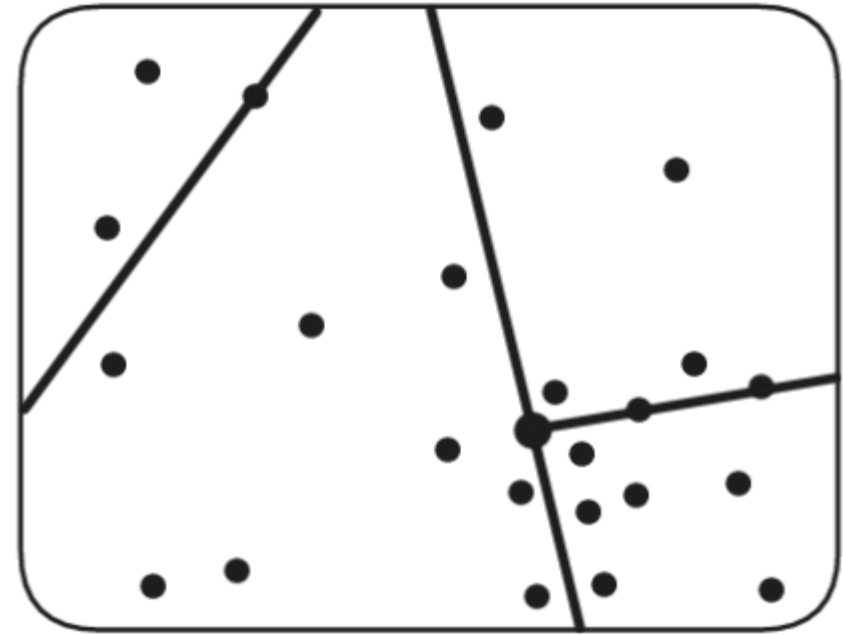


Song Structure around sequences

Generalization of buildings



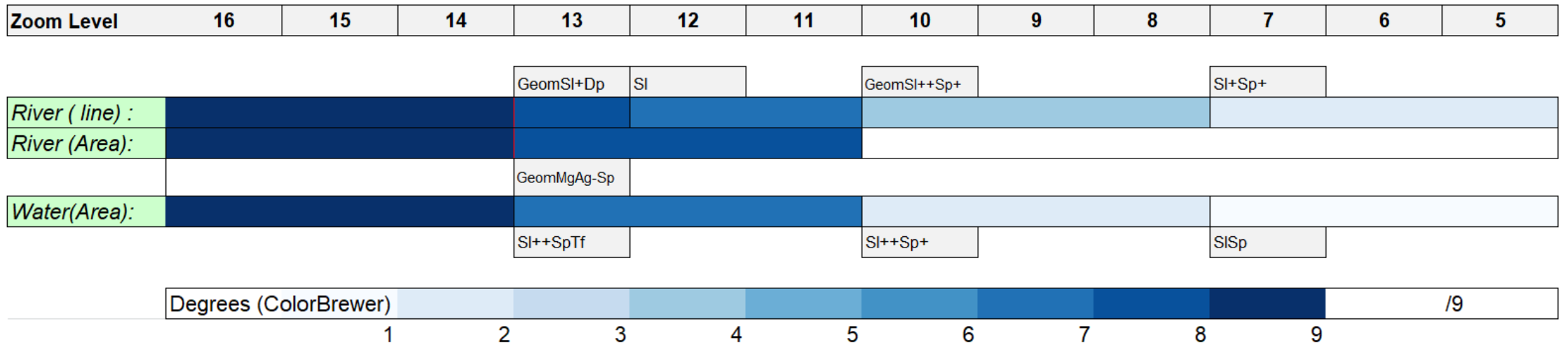
Finger-shaped form



Dispersed form

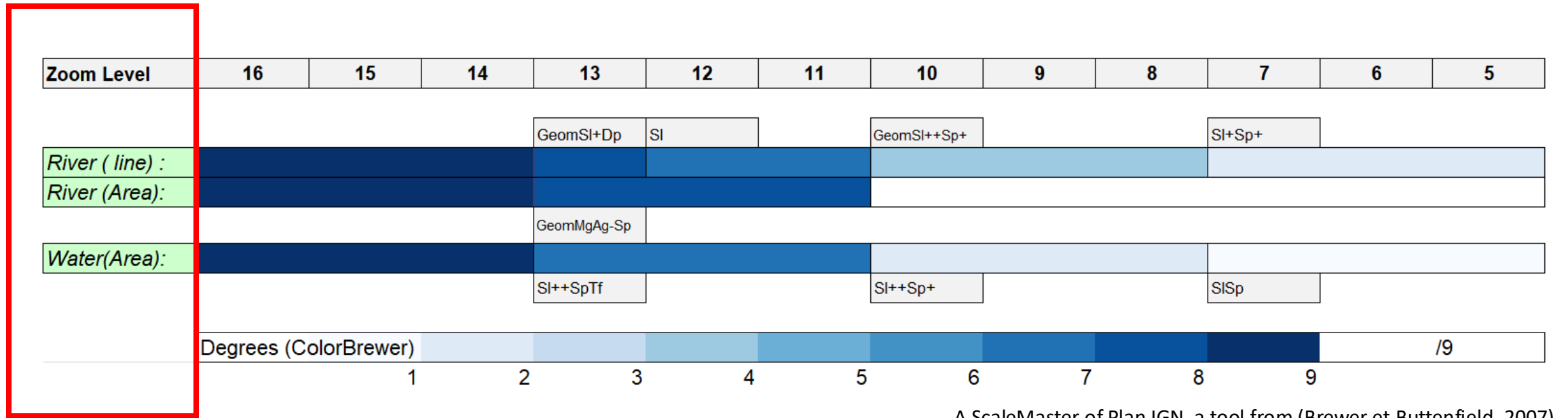
➡ Need of manual corrections for different patterns !

A ScaleMaster : a tool to illustrate sequences & rhythm



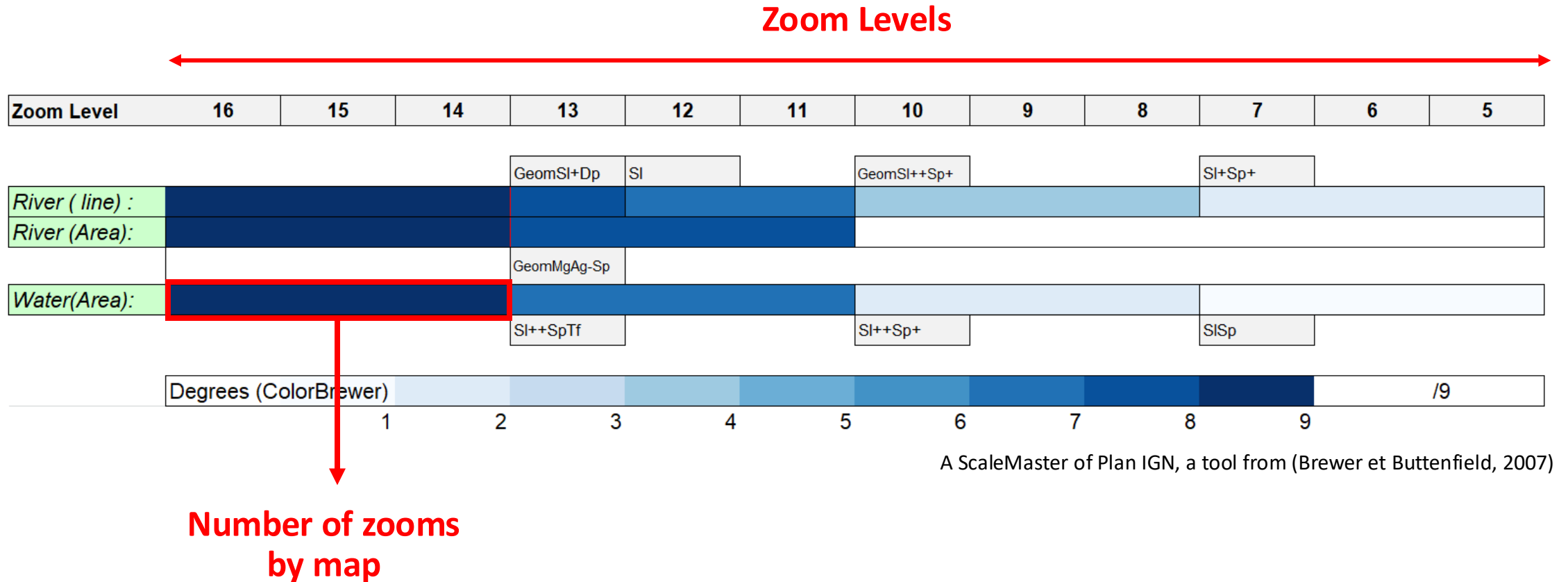
A ScaleMaster of Plan IGN, a tool from (Brewer et Buttenfield, 2007)

A ScaleMaster : a tool to illustrate sequences & rhythm

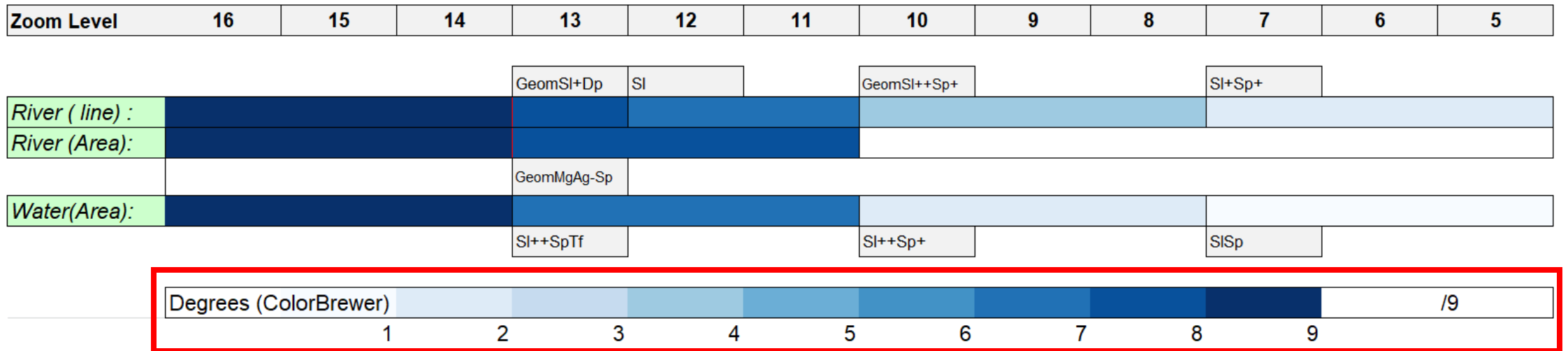


Data

A ScaleMaster : a tool to illustrate sequences & rhythm



A ScaleMaster : a tool to illustrate sequences & rhythm



Gap of abstraction

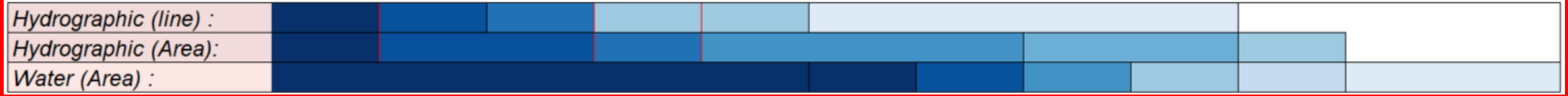
A ScaleMaster of Plan IGN, a tool from (Brewer et Buttenfield, 2007)

A ScaleMaster : a tool to illustrate sequences & rhythm

Main melody

[illegible]

OSM



Plan IGN



Two different ScaleMasters, with OpenStreetMap (pink) encouraging a successive display of its data, contrary to Plan IGN (green) that favours simultaneity

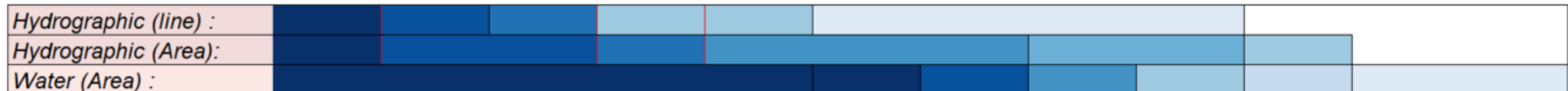
A ScaleMaster : a tool to illustrate sequences & rhythm

Fl. *piu tranquillo*
Ob.
Cl. (A)
Fag.



Harmonic
chord

OSM



Plan
IGN



Two different ScaleMasters, with OpenStreetMap (pink) encouraging a successive display of its data, contrary to Plan IGN (green) that favours simultaneity

Sequences in music

Les Beaux-Arts mystérieux

Violons II

Edouard Delale

Andante ♩ = 92

mf

5

10

p

A Poco capriccioso ♩ = 104

pizz.

mp

poco rit.

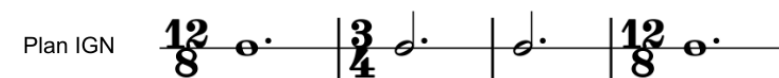
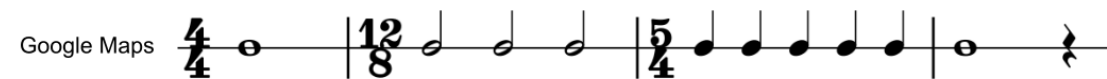
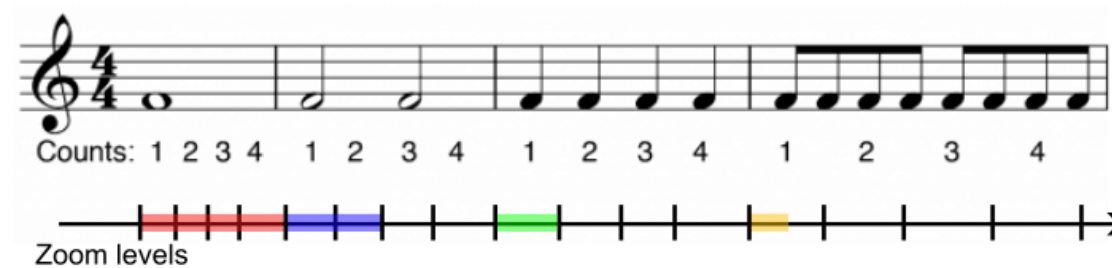
a tempo

mp

17

poco rit.

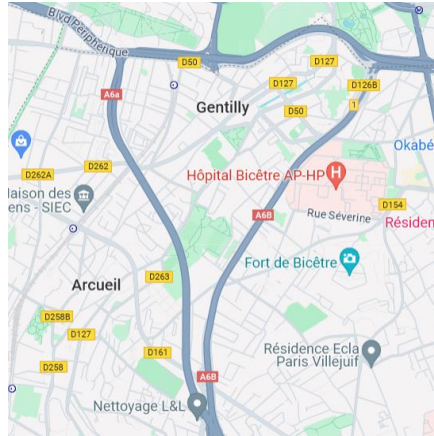
a tempo



Is Google Maps progressive ?



Wrong layering

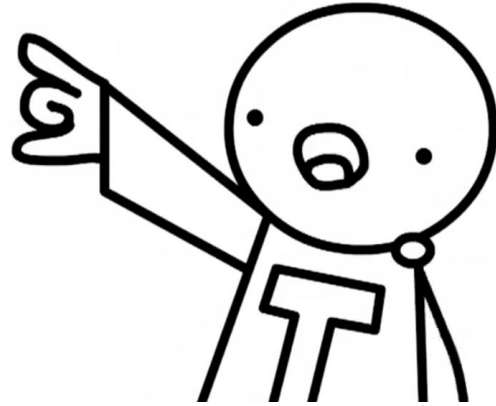


Inconsistent holes

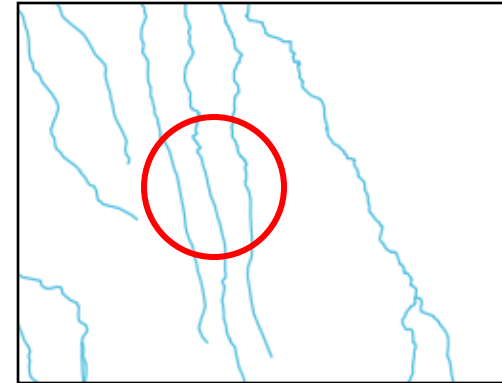
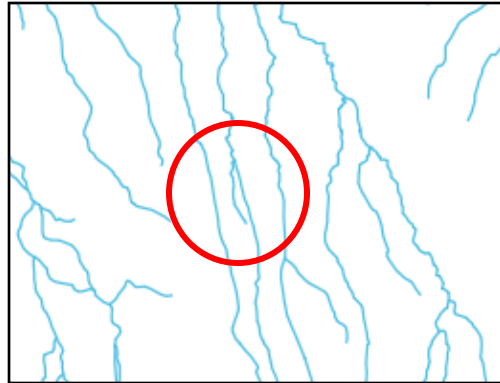
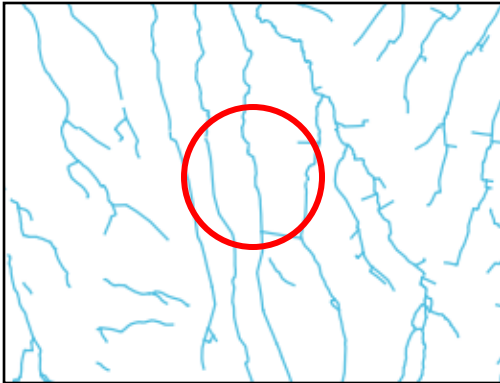


(Too ?) Fast rhythm

Handling consistency



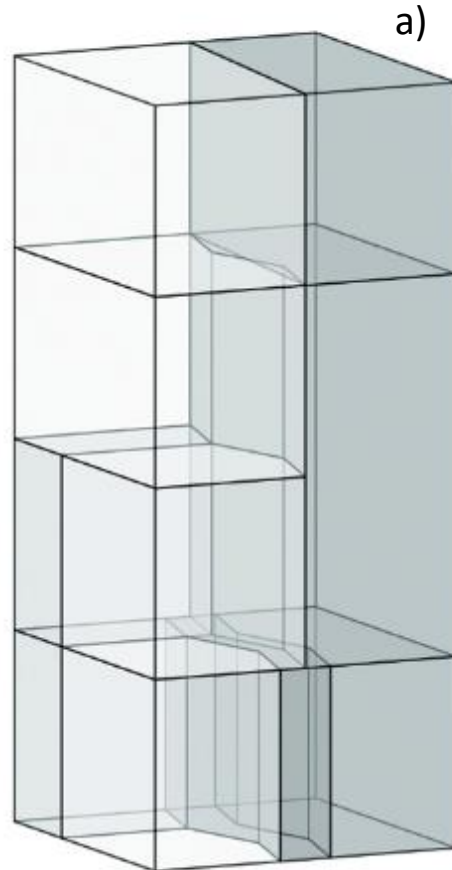
"Asdf movie" by Thomas James Ridgewell built on unpredictability



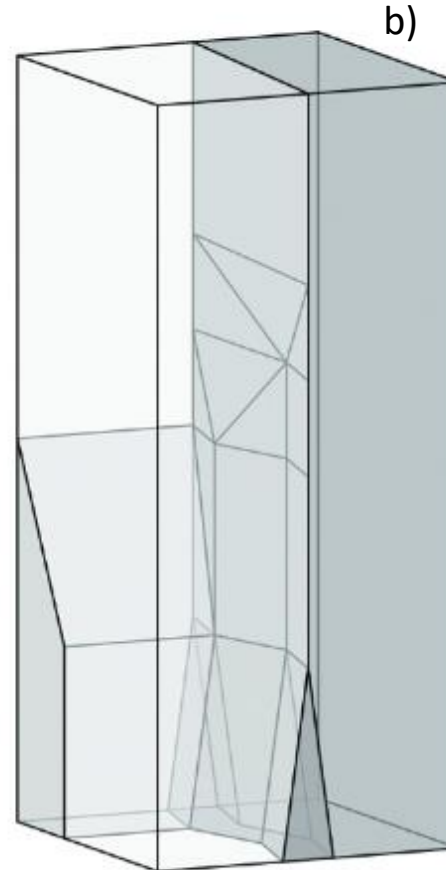
Predictability

Continuity VS progressiveness

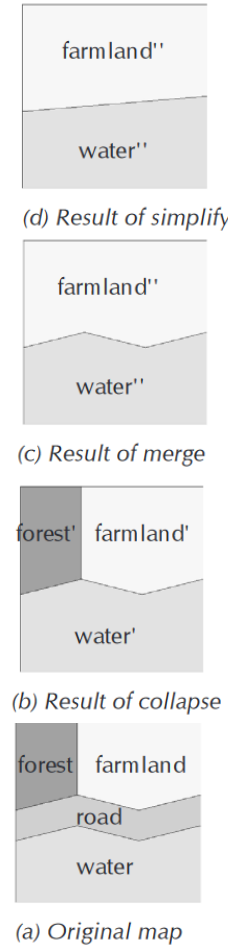
**Discrete
generalisation :**



**Continuous
generalisation :**



The space-scale cube representation in 3D with a) classic tGAP structure and b) a smooth tGAP structure (van Oosterom & Meijers, 2014)



Continuity VS progressiveness



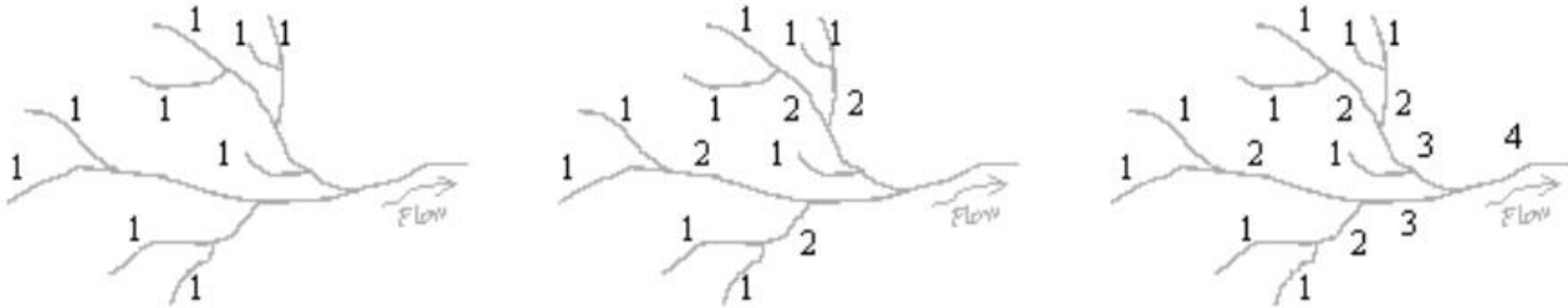
An example of a continuous generalisation disregarding sequences in a pan-scalar map of Limburg, by (Peng et al., 2023)

Between breaks and closure



"Mise au monde" (©Alvaro Nofuentes)

River network selection

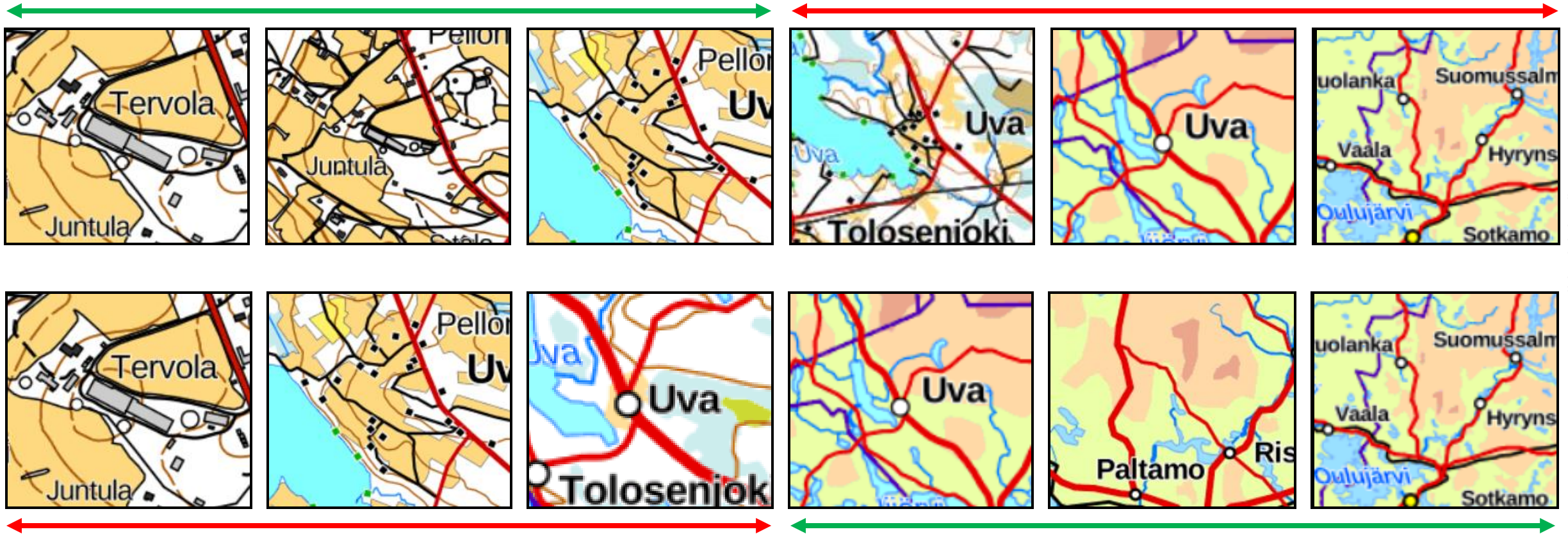


First stage of Horton ordering (Strahler ordering)



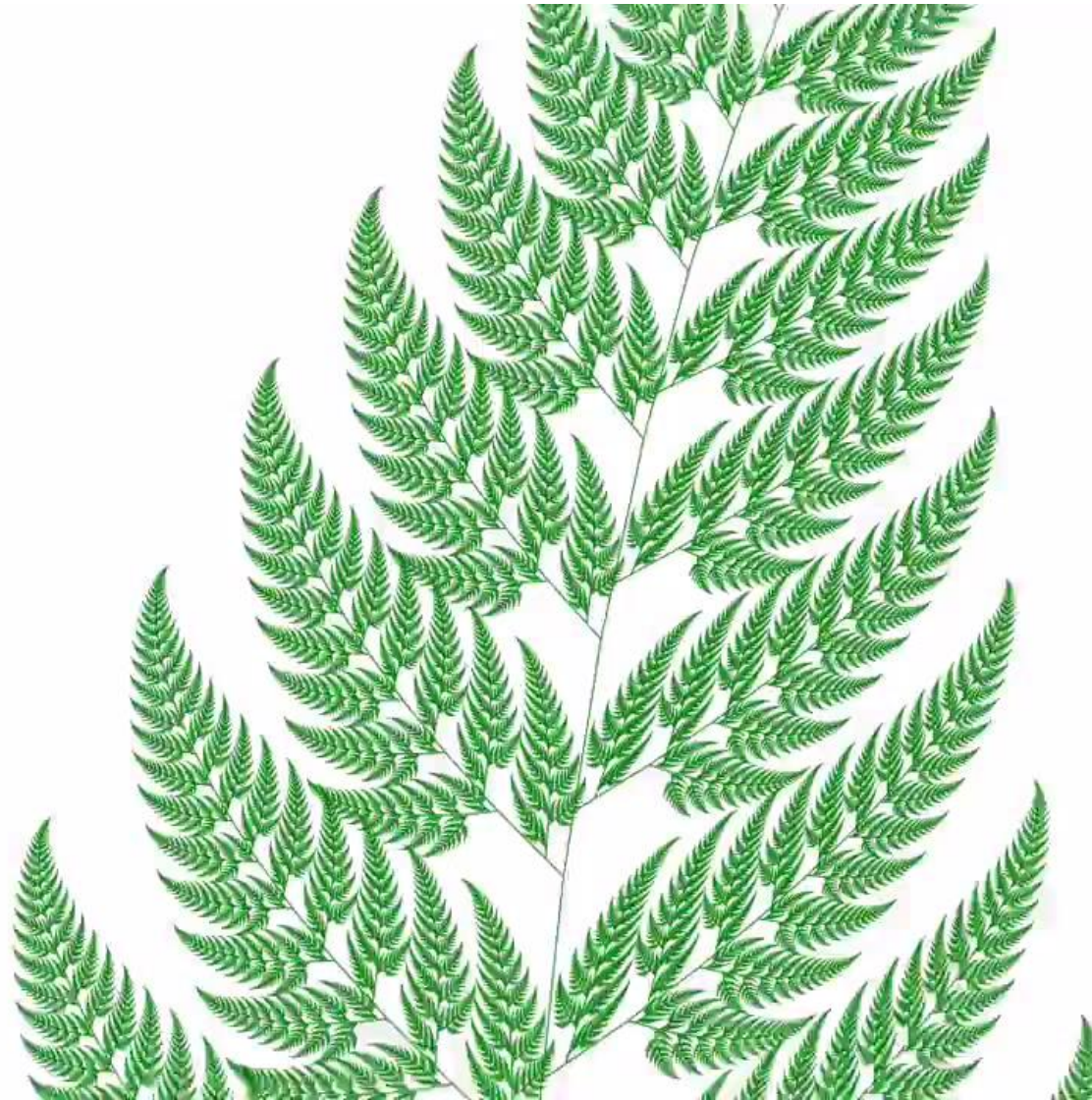
Second stage of Horton ordering - main streams defined

Handling rhythm



- Fast abstraction rhythm
- Slow abstraction rhythm

Fractals : a « pan-scalar » pattern

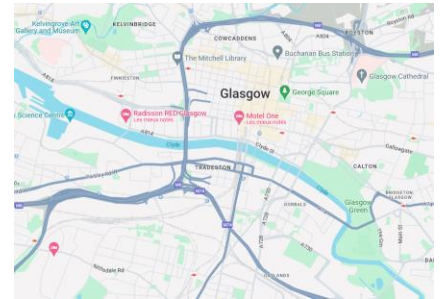
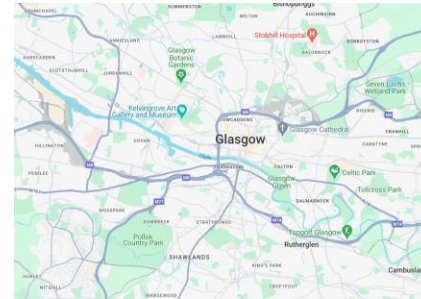
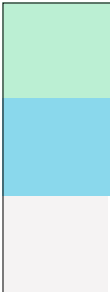


Barnsley Fern infinite zoom

Color consistency



7 rings – Ariana Grande



Google Maps



Plan IGN

Better understanding sequences

4
5
6
7
8
9
10
11
12
13
14
15
16
17
18

Zoom levels

1st sequence



Topographic
map

2nd sequence



Road
network

3rd sequence

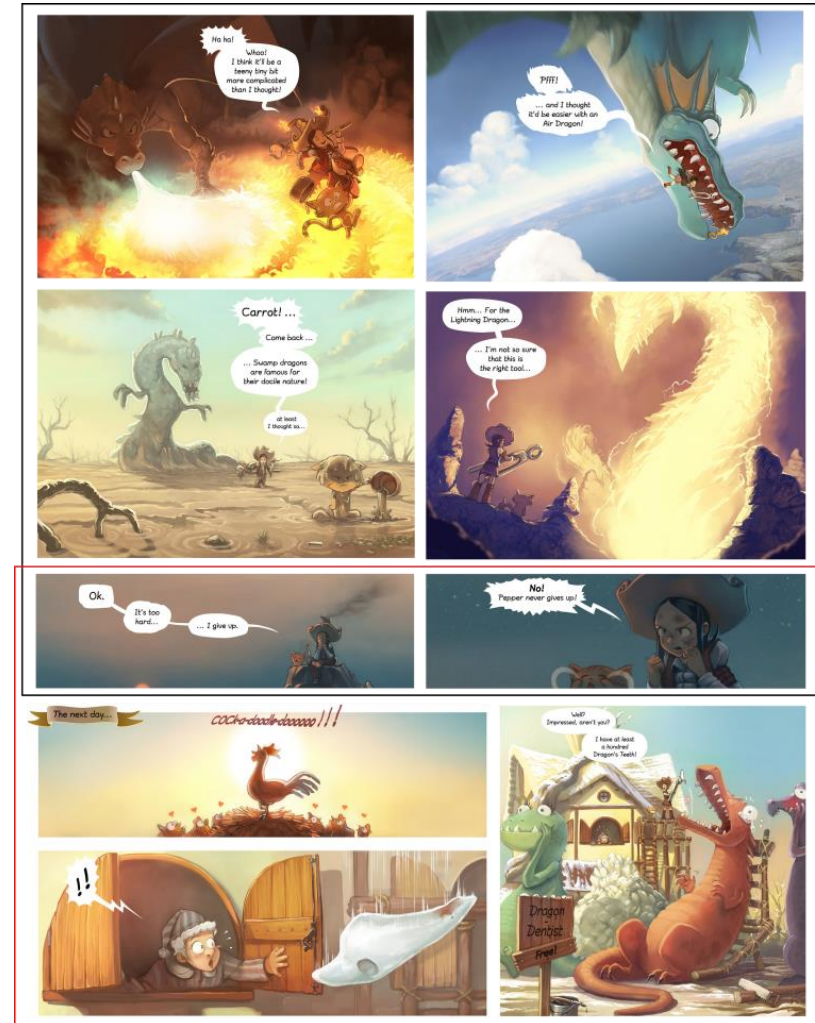


City
area

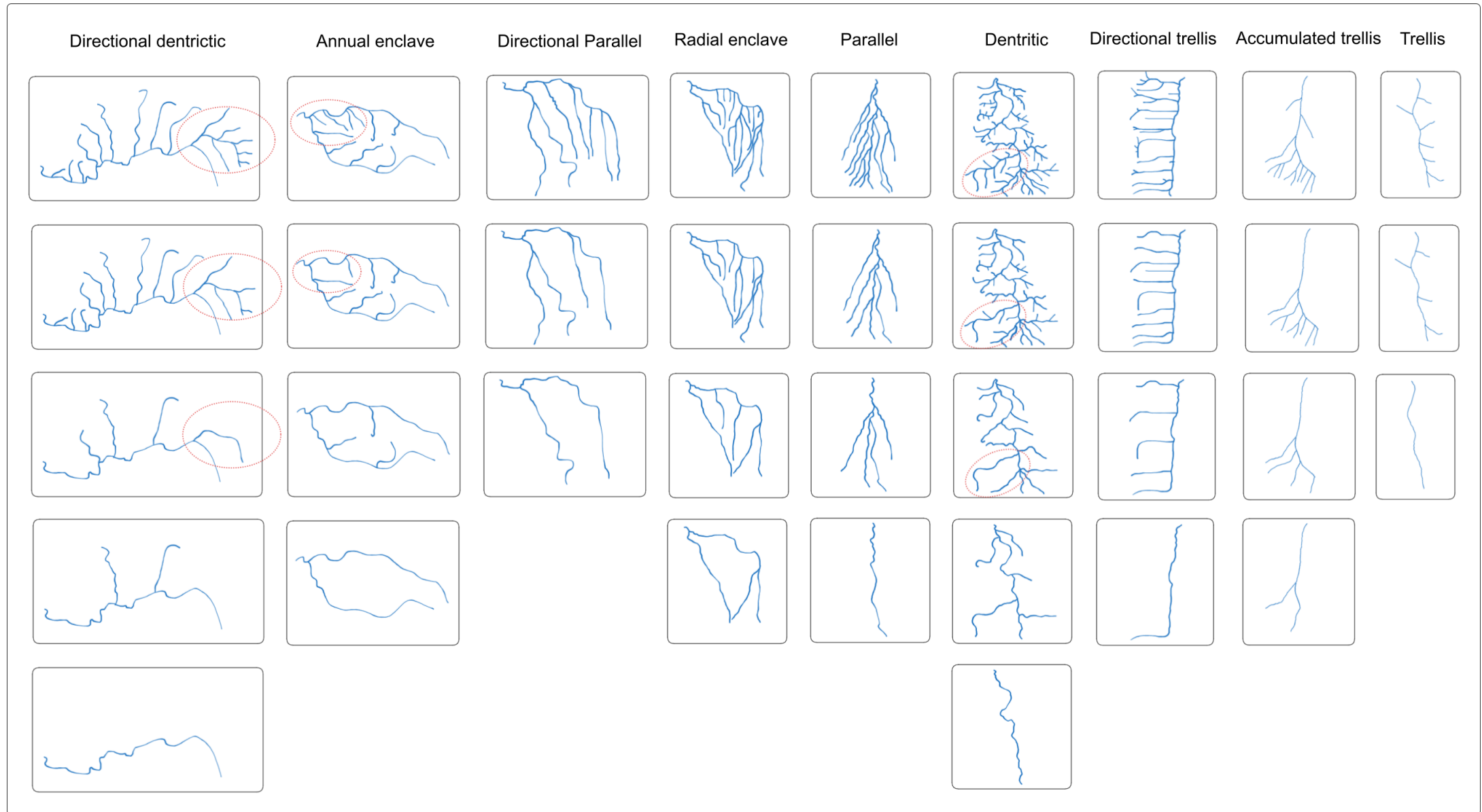
4th sequence



Buildings



Generalisation of each pattern



A pan-scalar organization of patterns

